Iosif's network duality is the Petri-net reverse dual: what it transfers and what it does not

chemistry-open-problems · 25 October 2026, rewritten 28 October 2026 · partial result

Claude Opus 5.5: research, literature, verification, writing · GPT-6-Astra: research, verification of the archived report

We thank Alexandru Iosif for helpful correspondence, which shaped questions Q1–Q3 and pointed us to recent work on the Global Attractor Conjecture.

Abstract. A. Iosif (arXiv:2603.08767) defines the dual $\mathcal{N}^*$ of a mass-action network $\mathcal{N}$ by exchanging species and rate constants, conjectures a "duality relation" between preclusters and maximal invariant polyhedral supports (Conjecture 22), asks for one between preclusters and siphons (Question 24), and hopes to transfer known cases of the Global Attractor Conjecture (GAC) "through duality, in small codimensions". Every precise form of the conjecture and of the question fails; we condense the exact criteria and the minimal counterexamples of an earlier report. We then answer the three questions behind them. (1) Iosif's dual is the weighted reverse dual of Petri-net theory (Hack 1972). Through it, siphons of $\mathcal{N}$ are the T-traps of $\mathcal{N}^*$, and for equal-conflict networks the seeds of preclusters are the Petri clusters and minimal siphons cross every cluster at most once, so preclusters and siphons are related by transversality, not refinement. Definition 19 depends on how a choice is split; the Conradi–Flockerzi clusters repair it, and for them maximal invariant polyhedral supports refine preclusters for every consistent (respectively conservative) network, with equality exactly when doubling pairs (respectively twin species) are coupled, in particular on the species side of every weakly reversible network satisfying the hypotheses of Conjecture 22. (2) The duality transfers structure on a self-dual class: if $\mathcal{N}$ is connected, conservative, consistent and satisfies the cluster rank equation $s = q - 1$ (the rank theorem of free-choice and equal-conflict nets), then neither $\mathcal{N}$ nor $\mathcal{N}^*$ has a critical siphon and both are persistent for every rate vector; for proportional-conflict networks both moreover have exactly one exponentially stable positive steady state in each class; for equal-conflict networks $\mathcal{N}$ is globally stable, and $\mathcal{N}^*$ too when $s \leq 2$. An explicit network outside proportional conflict has three steady states in one class. (3) Outside this class the duality transfers, in general, none of weak reversibility, endotacticity, the siphon criterion of Angeli, De Leenheer and Sontag, persistence (the dual of $2X_2 \rightleftarrows X_1 + X_2$ loses a species for every rate vector) and convergence (a weakly reversible deficiency-zero network has a dual that is persistent for all rates, with a unique positive equilibrium in every class, and, for some rate vector, a periodic orbit). The duality preserves the stoichiometric dimension, so it cannot turn small codimension into small dimension. For rank one we decide persistence of the dual exactly, with an all-or-none dichotomy for weakly reversible networks; in rank two the dichotomy is open. The GAC is now claimed in full by two preprints, which we have not checked. Every example is certified in exact arithmetic (130 claims); counts over large families are cited from the working notes as checked by computation. The main results were re-derived by three fresh-context verifiers.

1 Introduction

1.1 The questions

Iosif [1] develops a duality for mass-action networks. The dual $\mathcal{N}^*$ of a network $\mathcal{N}$ has one species for each reaction of $\mathcal{N}$ and one reaction for each species of $\mathcal{N}$; in his words the duality is between chemical species and rate constants. He proves that conserved quantities are dual to internal cycles (his Theorem 21) and then states, in §3 (p. 7 of arXiv v2, 15 July 2026; identical in v1 and in HAL v3):

> *"Conjecture 22. For conservative mass-action networks that do not have two species with exactly the same rates and have at least one positive steady state, there is a duality relation between the sets of preclusters and of maximal invariant polyhedral supports.*
> *Remark 23. The condition $\dot{x}_i \neq \dot{x}_j$ for $i \neq j$ is necessary in order to make sure no further clustering on the left kernel is done. We require that there is at least one positive steady state so that the internal cycles are a refinement of preclusters.*
> *We conclude with the following question:*
> *Question 24. Is there a duality relation between preclusters and siphons?"*

The paragraph that follows gives the motivation (p. 7–8):

> *"On the other hand, it would be interesting to understand whether a possible duality between preclusters and siphons would have consequences for toric dynamical systems. It is well known that the Global Attractor Conjecture holds in small dimensions [4], and we wonder whether there is a similar argument, through duality, in small codimensions."*

The source's reference [4] is Craciun, Dickenstein, Shiu and Sturmfels [2]. The abstract adds: "Given the close relation between maximal invariant polyhedral supports and siphons, we also conjecture that siphons and preclusters are dual objects." The statement has changed status over time: the 2020 HAL version states it as a theorem without proof [3], a 2024 talk lists it as a theorem [4], a 2025 poster abstract says it is proved [5], and the 2026 preprint calls it a conjecture. The source does not define "duality relation"; its closing diagram writes "Refine" both literally and in quotation marks. An earlier report of this project (archived as `archive/paper-precise-forms.pdf`, [6]) made each ingredient precise and refuted every precise form. Since the relation is left open, we ask three questions that the source supports:

Q1. Do preclusters and maximal invariant polyhedral supports (and siphons) correspond in a looser sense than equality, refinement or equal cardinality?

Q2. Is there a meaningful duality, and what does it transfer?

Q3. Can it transfer known cases of the GAC, or properties of the GAC setting?

1.2 The answers in one page

The dual is classical. Read as a Petri net (places = species, transitions = reactions, arc weights = stoichiometric coefficients), $\mathcal{N}^*$ is the weighted *reverse dual*: places and transitions are exchanged and every arc is reversed (Proposition 3). This construction goes back to Hack [7]. It yields a dictionary (Table 6): conservation laws of $\mathcal{N}$ are flux vectors of $\mathcal{N}^*$, siphons of $\mathcal{N}$ are the T-traps of $\mathcal{N}^*$, and the doubling pairs that seed Iosif's preclusters are the reaction halves of the Petri clusters when $\mathcal{N}$ is equal-conflict, every species is consumed and no source is zero (e.g. for connected networks with (H1) and (H2)).

Q1: transversality, and an invariant repair. Every precise form fails (Theorem 11). The reason on the siphon side is structural: a minimal siphon contains at most one species of each class of species with the same consumers (Theorem 20), so a precluster with two elements never lies in a minimal siphon (Theorem 21). Definition 19 depends on how a choice is split into binary choices (Proposition 12). The Conradi–Flockerzi clusters $\widehat{\mathsf{P}}$ are invariant under duplication and splitting (Theorem 16). For them the maximal invariant polyhedral supports refine preclusters on both sides, for every consistent (respectively conservative) network, and equality holds exactly when the twin (doubling) pairs are coupled, in particular on the species side of every weakly reversible network satisfying the hypotheses of Conjecture 22 (Theorem 19). This equality is close to built in, and the content of $\widehat{\mathsf{P}}$ is dynamical: Remark 23 for $\widehat{\mathsf{P}}$ is the isolation property that gives toric steady states (Proposition 18).

Q2: a self-dual class where structure transfers. The classical rank theorem of free-choice nets is a self-dual condition: connected, conservative, consistent, and $s = q - 1$, where q is the number of Petri clusters. Under it neither $\mathcal{N}$ nor $\mathcal{N}^*$ has a critical siphon, so both are persistent for every rate vector (Theorem 31; the ordinary-net siphon statement is due to Desel). For proportional-conflict networks, a self-dual subclass, both sides moreover have positive steady states for all rates, exactly one per class, and a Hurwitz Jacobian on the stoichiometric subspace at every positive point (Theorem 42). For equal-conflict networks the steady states of $\mathcal{N}$ are toric and globally attracting (Theorems 33, 35 and 37), and those of $\mathcal{N}^*$ are globally attracting when $s \leq 2$ (Proposition 40). Proportional conflict is the boundary of the dynamical statements: a network with the cluster rank equation but not proportional conflict has three positive steady states in one class, one a saddle (Proposition 44). Figure 46 summarises the classes.

Q3: what does not transfer. Outside these classes the dual leaves the GAC setting (Section 5): weak reversibility, deficiency, endotacticity and the siphon criterion of Angeli, De Leenheer and Sontag [8] are not preserved; persistence is not preserved, with extinction for every rate vector (Proposition 51 and Theorem 52); convergence is not preserved even when persistence is (Theorem 54). For rank one, persistence of the dual is decided exactly and is all-or-none for weakly reversible $\mathcal{N}$ (Theorems 58 and 60); in rank two the all-or-none question is open (Section 6.3). The duality keeps the stoichiometric dimension s and exchanges the codimension $n - s$ with the cycle-space dimension of $\mathcal{N}^*$, so there is no "small codimension" argument of the kind hoped for (Section 7). The GAC itself is now claimed in full by two preprints [9] [10], which we have not checked.

1.3 Results, status and credits

Table 1 lists the results. *Status*: proved (here, with a complete proof or a proof sketch marked as such; for the statements given without proof, in the working notes of Table 76, re-derived by the verifier), checked (exact computation on finite families, not a proof), refuted, or open. *Verification*: the fresh-context verifier that re-derived the result (V1, V2, V3, Section 9) or the verifier of the archived report. *Novelty*: "known" with the source when the statement is in print; "extends" when a special case is in print; "not found" when the literature search of Section 9.3 found no prior statement; this is not a claim of priority. The working notes behind each result are listed in the concordance

(Table 76).

Table 1 / Results of the paper. P.: proved; Ch.: checked by computation; R.: refuted.

No.	Statement	Status	Verif.	Novelty
3	Iosif's dual is the weighted reverse dual; dictionary	P.	V1	construction known (Hack; Desel–Esparza; Teruel–Silva); Γ^{T} networks known (Famili–Palsson; Blanchini–Giordano; Duvall); identification not found
11	every precise form of Conj. 22, Q. 24, Rem. 23 fails; exact criteria	P.	archive	not found
12	Def. 19 depends on splitting and duplication	P.	V1	not found
16	Conradi–Flockerzi clusters $\widehat{\mathsf{P}}$ are invariant; full coupling is forced	P.	V1	not found
17	$\widehat{\mathsf{P}}$ is one block for connected EC networks; linkage classes for WR deficiency 0	P.	V1	not found
18	Remark 23 for $\widehat{\mathsf{P}}$ = isolation $\Rightarrow$ toric for all rates	P.	V1	extends Iosif's thesis Thm 4.17
19	$\mathsf{M} \preceq \widehat{\mathsf{P}}$ on both sides; equality iff doubling (twin) pairs coupled	P.	V1 (corr.)	not found
20	transversality of minimal siphons	P.	V1	extends Desel–Esparza Thm 4.30(a)
21	exact criterion for "preclusters lie in minimal siphons"	P.	V1 (corr.)	not found
24	structure of well-formed free-choice networks	P. (sketch)	V1	assembled from Desel–Esparza
25	no siphon-based description of P outside free choice	Ch.	V1, V2	not found
31	cluster rank equation: no critical siphon on either side; persistence	P.	V2	ordinary nets: Desel 1994; weighted: not found
33	WF-EC: positive toric steady states for all rates iff $s = m - 1$	P.	V2	extends Pérez Millán et al. 2012, Campos et al. 1991; cf. Recalde et al. 2002, Júlvez et al. 2005
35	WF-EC: one steady state per class; Hurwitz at every point	P.	V2	not found (tools classical)
37	WF-EC: global stability	P.	V2	not found (technique classical)

No.	Statement	Status	Verif.	Novelty
38, 39	dual of WF-EC: existence for all rates, uniqueness, Hurwitz	P.	V2	not found
40	dual of WF-EC globally stable if $s \leq 2$	P.	V2	not found
42	proportional conflict: self-dual; existence, uniqueness, stability, persistence	P.	V2	not found
44	cluster rank equation does not give uniqueness (NC3)	P.	found by V2; certificate	not found
47	WR $\Rightarrow$ dual has equilibria for all rates; equilibrium manifold	P.	V2, V3	convex argument classical (Birch)
48, 49	WR, deficiency, ADLS criterion not preserved	P.	V3	WR: known (Blanchini–Giordano); rest not found
51, 52	extinction of the dual for every rate vector (drain species, drain sets)	P.	V3	not found
54	persistent dual with a unique equilibrium per class and a periodic orbit	P.	V3	not found
58, 60	rank one: exact persistence test; all-or-none for WR	P.	V3 (corr.)	"if" half of (1) under a stronger hypothesis: Zhang–Gao–Dochain 2023; rest not found
63	WR: no dominated facet of the dual, any rank	P.	V3	not found
65, 66	rank two: persistence certificates; counts	P. (sketch), Ch.	V3 (corr.)	tools classical
68, 70	a persistent dual with an OUT facet (refutes a conjecture of agents Q1 and D2); the family $N_{u,R}$	P.; P. (sketch)	V3	not found
72, 73	duality keeps s; prime-cycle certificate	P.; P. (sketch)	V3	ingredients classical
74	boundary conditions of GAC theorems in dual terms	P.	V3	re-expression of ADLS, Anderson–Shiu, Wiuf

2 Setting and the Petri dictionary

2.1 Networks, hypotheses and the dual

A *network* $\mathcal{N}$ has species $X_1, \dots, X_n$ and reactions $R_j : y_j \to y'_j$ $(j = 1, \dots, r)$ with complexes $y_j, y'_j \in \mathbb{Z}^n_{\geq 0}$. The educt and product matrices are $Y_{\mathrm{e}} = (y_1 | \cdots | y_r)$ and $Y_{\mathrm{p}} = (y'_1 | \cdots | y'_r)$, the stoichiometric matrix is $\Gamma = Y_{\mathrm{p}} - Y_{\mathrm{e}}$, $s = \operatorname{rank} \Gamma$, and $S = \operatorname{im} \Gamma$ is the stoichiometric subspace. The mass-action system with rate vector $k \in \mathbb{R}^r_{>0}$ is $\dot{x} = \Gamma\,(k_j x^{y_j})_j$. A *conservation law* is z with $z^{\mathsf{T}}\Gamma = 0$; a *flux vector* is $v \in \ker \Gamma$.

Compatibility classes are $(x^0 + S) \cap \mathbb{R}^n_{\geq 0}$. The complexes and reactions form a directed graph whose connected components are the *linkage classes*; $\mathcal{N}$ is *weakly reversible* if every reaction lies on a directed cycle of this graph, and its *deficiency* is the number of complexes minus the number of linkage classes minus s [11]; complex balance and (strong) endotacticity are used as in [12] [13] [14]. Conjecture 22 assumes:

(H1) $\mathcal{N}$ is *conservative*: some conservation law is strictly positive;

(H2) $\mathcal{N}$ has a positive steady state for some k, equivalently it is *consistent*: some $v \in \ker \Gamma$ is strictly positive (if $x > 0$ is a steady state, $v = (k_j x^{y_j})_j$ works; conversely $x = (1, \ldots, 1)$ is a steady state for $k = v$);

(H3) no two species have the same rate function: the rows of Γ are pairwise distinct.

A *siphon* is a nonempty set Σ of species such that every reaction producing a species of Σ consumes one (Definition 13 of the source; [15] [8]); a *trap* is defined with production and consumption exchanged. The support of a nonzero conservation law $z \geq 0$ is a siphon. A siphon is *critical* if it contains the support of no such law. A system is *persistent* if no trajectory starting in $\mathbb{R}^n_{>0}$ has an ω-limit point on the boundary (all trajectories below are bounded). For the dual we say ALL, SOME or NONE when it is persistent for every, for some but not every, or for no rate vector.

Following §3 of the source (see also [16]), the *dual* $\mathcal{N}^*$ has species $K_1, \ldots, K_r$, one for each reaction of $\mathcal{N}$, and one reaction $\rho_i \colon \sum_j (Y_\mathrm{e})_{ij} K_j \to \sum_j (Y_\mathrm{p})_{ij} K_j$ for each species X_i, with rate constant κ_i. In Iosif's reading the rate constants of $\mathcal{N}$ evolve and the concentrations of $\mathcal{N}$ play the role of rate constants. A pure catalyst of $\mathcal{N}$ gives a loop $y \to y$ of $\mathcal{N}^*$, and a species never consumed gives the zero complex; all definitions apply to such networks. Networks with stoichiometric matrix $\pm\Gamma^\mathsf{T}$ appeared before: Famili and Palsson's compound map of $-\Gamma^\mathsf{T}$, on which the extreme conservation laws are drawn as extreme pathways [17]; the dual network of Blanchini and Giordano [16], developed at the same time as Iosif's (source, p. 8); and Duvall's dual of an irreversible network [18]. Jamshidi and Palsson compare the Jacobians ΓG and $G\Gamma$ [19]. These works transfer linear and local properties (conservation pools, spectra, structural stability, monotonicity), not siphons or persistence.

Lemma 2 (transposition Lemma 2 [6]). $Y_\mathrm{e}(\mathcal{N}^*) = Y_\mathrm{e}^\mathsf{T}$, $Y_\mathrm{p}(\mathcal{N}^*) = Y_\mathrm{p}^\mathsf{T}$, $\Gamma(\mathcal{N}^*) = \Gamma^\mathsf{T}$ and $\mathcal{N}^{**} = \mathcal{N}$. Hence $\mathcal{N}^*$ has r species, n reactions and rank s; $\mathcal{N}$ satisfies (H1) iff $\mathcal{N}^*$ satisfies (H2), and conversely; $\mathcal{N}^*$ satisfies (H3) iff the columns of Γ are pairwise distinct; the internal cycles of $\mathcal{N}^*$ (supports of extreme flux rays) are the supports of the extreme conservation laws of $\mathcal{N}$ (Theorem 21 of the source).

2.2 The reverse dual

The species–reaction Petri net of $\mathcal{N}$ [8] has places X_i, transitions R_j, an arc $X_i \to R_j$ of weight $(Y_\mathrm{e})_{ij}$ and an arc $R_j \to X_i$ of weight $(Y_\mathrm{p})_{ij}$. For a transition t, ${}^\bullet t$ and $t^\bullet$ are its input and output places; for a place p, ${}^\bullet p$ and $p^\bullet$ are the transitions producing and consuming it. The *reverse dual* of a net (P, T, F) is (T, P, F^{-1}): places and transitions exchanged, every arc reversed (Hack [7]; Desel and Esparza Def. 6.19 [20], who note that it is sometimes called the reverse dual; Teruel and Silva §2 [21] for weighted nets, with incidence matrix C^T; Jančar, Best, Devillers and Ošťádal [22]).

Proposition 3 (the dual is the weighted reverse dual)**.** The Petri net of $\mathcal{N}^*$ is the reverse dual of the Petri net of $\mathcal{N}$, with every weight kept.

Proof. ρ_i consumes K_j with coefficient $(Y_\mathrm{e})_{ij}$ and produces it with coefficient $(Y_\mathrm{p})_{ij}$. So the arc $X_i \to R_j$ of weight $(Y_\mathrm{e})_{ij}$ becomes the arc from the place K_j to the transition ρ_i with the same weight, and the arc $R_j \to X_i$ becomes the arc $\rho_i \to K_j$. □

The identification is definitional, but we found it stated nowhere: no paper on the dual network mentions Petri reverse duals, and no Petri-net paper mentions network duals (Section 9.3). Figure 4 shows it on the network $3A \rightleftarrows 2A + B$ used in Section 5.

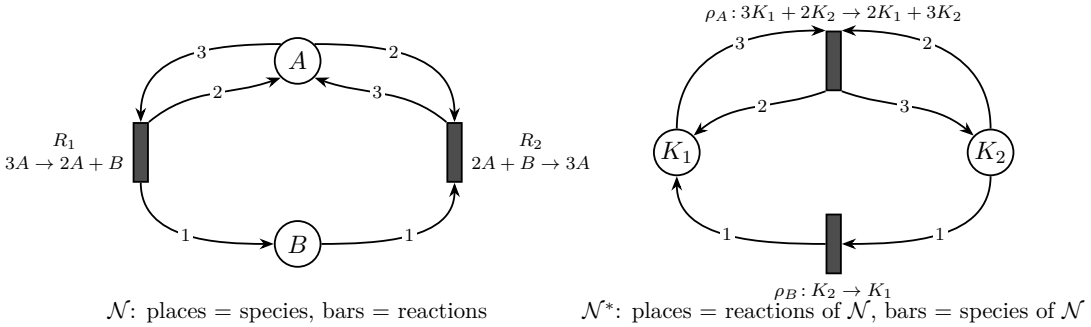


Figure 4 / The network $3A \rightleftarrows 2A + B$ and its dual as weighted Petri nets. Every arc of $\mathcal{N}$ is reversed and its weight kept; places and transitions are exchanged. The matrices are $Y_\mathrm{e} = \left(\begin{smallmatrix} 3 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, $Y_\mathrm{p} = \left(\begin{smallmatrix} 2 & 3 \\ 1 & 0 \end{smallmatrix}\right)$ for $\mathcal{N}$ and their transposes for $\mathcal{N}^*$; both networks have $\Gamma = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. $\mathcal{N}$ is reversible and globally stable; $\mathcal{N}^*$ is not weakly reversible and loses K_2 for every rate vector (Example 50).

Petri vocabulary for reaction networks. A *Petri cluster* is a connected component $c = (S_c, T_c)$ of the bipartite graph with an edge $\{X_i, R_j\}$ whenever $(Y_\mathrm{e})_{ij} > 0$ (Desel–Esparza Def. 4.4); q is their number. Two reactions with the same source form a *doubling pair*; the classes of reactions with equal nonzero source are the *doubling classes* (Teruel and Silva's equal-conflict sets). Two species with equal rows of Y_e, zero rows included, are *twins*; the twin pairs of $\mathcal{N}$ are the doubling pairs of $\mathcal{N}^*$ (two zero rows give two reactions of $\mathcal{N}^*$ with the zero source), and the twin classes of nonzero rows are its doubling classes. A *T-trap* is a nonempty set U of reactions such that every species produced by U is consumed by U; a *T-siphon* is defined with the inclusion reversed. $\mathcal{N}$ is *connected* if the bipartite graph of all nonzero entries of Y_e and Y_p is connected. S-components (strongly connected subnets generated by a set of places with all their input and output transitions, in which every transition has one input and one output place) and T-components (places and transitions exchanged) are as in Desel and Esparza Defs 5.1, 5.11 [20].

Definition 5 (classes of networks)**.** $\mathcal{N}$ is *equal-conflict* (EC) if two reactions sharing a reactant species have the same source; equivalently, the m distinct source complexes have pairwise disjoint supports. It is *free-choice* if it is EC and ordinary (all coefficients 0 or 1). It has *balanced joins* if every source has all its nonzero coefficients equal. It is *proportional-conflict* (PC) if two reactions sharing a reactant have positively proportional sources; equivalently, on each Petri cluster $Y_\mathrm{e}[S_c, T_c]$ is a rank-one matrix $b_c a_c^\mathsf{T}$ with b_c primitive. It is *block-homogeneous* (BH) if every

source is a positive multiple of the indicator of a block of a partition of the species. $\mathcal{N}$ satisfies the *cluster rank equation* if $s = q - 1$. It is *well-formed equal-conflict* (WF-EC) if it is EC, connected, satisfies (H1), (H2) and $s = m - 1$.

EC is PC with a_c constant on each cluster, and BH is PC with $b_c = \mathbf{1}$. Under (H2) every species that occurs is consumed (a zero row of Y_e gives a row of Γ that is ≥ 0 and annihilates a positive vector), and under (H1) a reaction with the zero source is $0 \to 0$ ($z^{\mathsf{T}} y' = 0$ for a positive law z); a species in no reaction, or a reaction $0 \to 0$, is isolated in the graph of Y_e and Y_p. So for connected EC networks with (H1) and (H2), clusters correspond to sources and $q = m$; with (H2) alone, $0 \to X_1$, $0 \to X_2$, $X_1 + X_2 \to 0$ is connected and EC with $m = 2$ and $q = 3$. WF-EC is defined algebraically. For pure nets (no species on both sides of a reaction) it is Petri well-formedness by the rank theorem of Teruel and Silva Thm 28 [21], who assume purity; for ordinary free-choice nets this is Desel and Esparza's Rank Theorem 6.14.

$\mathcal{N}$	$\mathcal{N}^*$	proof
species / reactions	reactions / species	definition
conservation law, (H1)	flux vector, (H2)	$\Gamma(\mathcal{N}^*) = \Gamma^{\mathsf{T}}$
(H3): rows of Γ distinct	columns of Γ distinct	same
siphon (trap)	T-trap (T-siphon)	below
critical siphon	T-trap containing no internal cycle	from the row above
Petri cluster (S_c, T_c)	Petri cluster (T_c, S_c)	same bipartite graph
twin pairs / doubling pairs	doubling pairs / twin pairs	definitions
free-choice; EC with balanced joins	free-choice; EC with balanced joins	$Y_e[S_c, T_c] = a_c J$ is symmetric; Teruel–Silva Cor. 32
PC; cluster rank equation	PC; cluster rank equation	Theorems 31 and 42
connected	connected	same graph
MIPS (Proposition 8)	classes of fully coupled reactions	Prop. 6 [6]
supports of extreme conservation laws	internal cycles	Theorem 21 of the source
well-formed free-choice	well-formed free-choice	Hack; Desel–Esparza Thm 6.21

Table 6 / The dictionary of the reverse dual. All rows follow from Lemma 2 and Proposition 3 by transposition, except those with a reference.

Siphons: the reactions ρ_i ($i \in \Sigma$) of $\mathcal{N}^*$ produce exactly the K_j with $(Y_p)_{ij} > 0$, i.e. the reactions of $\mathcal{N}$ that produce a species of Σ, and consume exactly the K_j with $(Y_e)_{ij} > 0$. So Σ is a T-trap of $\mathcal{N}^*$ iff Σ is a siphon of $\mathcal{N}$. Jančar et al. §4 [22] name the reverse-dual counterparts of minimal siphons "minimal T-traps" for ordinary free-choice nets; Hack [7] had no interpretation of the reverse dual of a trap. EC alone is not self-dual: $2X_1 + X_2 \rightleftarrows X_3$ is EC, and its dual $2K_1 \to 2K_2$, $K_1 \to K_2$, $K_2 \to K_1$ is not. Teruel and Silva p. 293 [21] make the same observation for weighted nets: equal-conflict nets with balanced joins form a self-reverse-dual class, and their Duality Theorem

Cor. 32 [21] states that a pure net is a well-formed balanced-joins EC net iff its reverse dual is.

Lemma 7 (strong connectivity; agent P)**.** If $\mathcal{N}$ is connected and satisfies (H1) and (H2), its Petri net is strongly connected; so is the *cluster graph* with an arc $c \to d$ when a reaction of T_c produces a species of S_d.

Proof. Take a terminal strong component K with places P_K and transitions T_K, and $z > 0, v > 0$ as in (H1), (H2). Then $0 = \sum_{p \in P_K} z_p(\Gamma v)_p$ splits into $\sum_{t \in T_K} v_t \sum_{p \in P_K} z_p \Gamma_{pt}$, whose inner sums are $\geq \sum_p z_p \Gamma_{pt} = 0$ because every product of $t \in T_K$ lies in P_K, plus $\sum_{t \notin T_K} v_t \sum_{p \in P_K} z_p (Y_{\mathrm{p}})_{pt} \geq 0$. Both vanish, so no arc enters K either, and K is everything by connectedness. A path of the Petri net maps to a path of the cluster graph. □

3 Preclusters, maximal invariant polyhedral supports and siphons

3.1 Definitions and the precise forms

Preclusters. Let E be an $r \times p$ matrix whose columns are the extreme rays of the flux cone $\ker \Gamma \cap \mathbb{R}^r_{\geq 0}$, with rows $n_1, \ldots, n_r$. The source's Definition 19 reads:

> *"Consider the graph $\mathfrak{G}$ over $[r]$ which has an edge $\{i, j\}$ whenever the ith and the jth arrows have the same source and let $n_1, \ldots, n_r$ denote the rows of a matrix with columns the rays of the stoichiometric cone. If to the graph $\mathfrak{G}$ we adjoin the edges $\{i, s\}$ and $\{j, s\}$ whenever n_s is in the span of $\{n_i, n_j\}$ for each edge $\{i, j\}$ of $\mathfrak{G}$, we obtain the preclustering graph. A precluster is a connected component of the preclustering graph."*

We write $\mathsf{P}(\mathcal{N})$ for the partition into preclusters. The iterated version of Iosif's thesis Def. 4.3, Alg. 4.4 [23] gives the same partition Lemma 5 [6]. The span rule comes from Conradi and Flockerzi Fact 2 [24]. Under (H2) all n_s are nonzero. Reactions s, t are *fully coupled* if $n_s \in \mathbb{R}_{>0} n_t$, i.e. v_s / v_t is constant on $\ker \Gamma$ (flux coupling analysis [25]; enzyme subsets [26]). The supports of the extreme flux rays are the internal cycles (Definition 20 of the source; elementary flux modes [27]).

MIPS. Definition 17 of the source and its Remark 18 define maximal invariant polyhedral supports through the decorated combinatorial types of the invariant polyhedra. The archived report proves that they are the following classes.

Proposition 8 (Prop. 3, Prop. 6 [6])**.** Assume (H1). Species i, j lie in one maximal invariant polyhedral support iff their columns w_i, w_j in the matrix of extreme conservation laws are positively proportional, iff $e_i - \mu e_j \in S$ for some $\mu > 0$. We write $\mathsf{M}(\mathcal{N})$ for this partition. If $\mathcal{N}$ satisfies (H2), $\mathsf{M}(\mathcal{N}^*)$ is the partition of the reactions of $\mathcal{N}$ into classes of fully coupled reactions.

Two species are *conservation-coupled* if they lie in one block of $\mathsf{M}(\mathcal{N})$. $A \preceq B$ means that every block of A lies in a block of B.

Theorem 9 (exact criteria Thm 7, Cor. 8–10 [6])**.** Let $\mathcal{N}$ satisfy (H2).

(1) $\mathsf{P}(\mathcal{N}) \preceq \mathsf{M}(\mathcal{N}^*)$ iff every doubling pair is fully coupled.

(2) $\mathsf{M}(\mathcal{N}^*) \preceq \mathsf{P}(\mathcal{N})$ iff for every fully coupled class κ with $|\kappa| \geq 2$ the line $\mathbb{R}n_\kappa$ lies in $\operatorname{span}\{n_i, n_j\}$ for some doubling pair $\{i, j\}$.

(3) $\mathsf{P}(\mathcal{N}) = \mathsf{M}(\mathcal{N}^*)$ iff every doubling pair is fully coupled and every fully coupled class with at least two elements contains a doubling pair.

The species side is the same statement for $\mathcal{N}^*$ under (H1), with twin pairs and conservation-coupled pairs. If (H1), (H3) hold and every product complex is a source complex (e.g. $\mathcal{N}$ weakly reversible), $\mathcal{N}$ has no twins, so $\mathsf{P}(\mathcal{N}^*)$ is discrete. If $\mathcal{N}$ is weakly reversible without loops and some complex is the source of two reactions, $\mathsf{P}(\mathcal{N}) \not\preceq \mathsf{M}(\mathcal{N}^*)$; at deficiency zero $\mathsf{P}(\mathcal{N}) \neq \mathsf{M}(\mathcal{N}^*)$ always.

Proof. Two nonzero nonnegative dependent vectors are positively proportional. (1) A doubling pair lies in one precluster; conversely, if all doubling pairs are fully coupled, $\operatorname{span}\{n_i, n_j\} = \mathbb{R}n_i$ for each, so every adjoined s is coupled to i. (2) and (3) follow the same way from the description of the edges of the preclustering graph. Twins: if X_i, X_j are twins and every product is a source, rows i, j of Y_p agree too, contradicting (H3). WR: a shortest path from the product of $R_a : y \to y_1$ back to y avoids $R_b : y \to y_2$, so the cycle through R_a gives a flux with $v_a = 1$, $v_b = 0$. At deficiency zero without branching, every linkage class is a cycle whose reactions are fully coupled, a class without a doubling pair. □

	Formulation	Fails for
A1–A4	$\mathsf{P}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$; $\mathsf{P}(\mathcal{N}^*) \preceq \mathsf{M}(\mathcal{N})$; $\mathsf{M}(\mathcal{N}) \preceq \mathsf{P}(\mathcal{N}^*)$; equal numbers of blocks	R, T, U; W^*, U; R, T; R, T
B1–B4	the same on the reaction side, $\mathsf{P}(\mathcal{N})$ versus $\mathsf{M}(\mathcal{N}^*)$	R, T, W, N_3; W, N_3; $R, T, N_{4.12}$; R, T
C	$\lvert\mathsf{P}(\mathcal{N})\rvert = \lvert\mathsf{M}(\mathcal{N})\rvert$	R, T
Q1–Q6	$\mathsf{P} = \mathsf{S}$; $\mathsf{S} \preceq \mathsf{P}$; $\mathsf{P} \preceq \mathsf{S}$; preclusters are siphons; unions of minimal siphons; $\lvert\mathsf{P}\rvert = \lvert\mathsf{S}\rvert$ (both sides)	T; T; D, F; T; T; T
R23	every internal cycle lies in a precluster	$T, N_{4.12}, R$

Table 10 / Precise forms of Conjecture 22, Question 24 and Remark 23, with counterexamples satisfying (H1)–(H3) (condensed from [6], where three readings of MIPS are treated). S = minimal siphons.

The networks are $R : X_1 \rightleftarrows X_2$; the 3-cycle $T : X_1 \to X_2 \to X_3 \to X_1$ (weakly reversible, deficiency zero, self-dual); $W : X_1 \to X_2,\ X_1 \to X_3,\ X_2 \to X_1,\ X_3 \to X_1$ and its dual $W^* : K_1 + K_2 \to K_3 + K_4,\ K_3 \to K_1,\ K_4 \to K_2$; $U : X_4 \to X_2 + X_3,\ X_3 + X_4 \to X_1,\ X_1 + X_2 \to 2X_4$; $D : X_1 \to X_2,\ X_1 \to X_3,\ X_2 + X_3 \to 2X_1$; the one-site futile cycle $F : S_0 + E \rightleftarrows ES_0 \to S_1 + E,\ S_1 + F \rightleftarrows FS_1 \to S_0 + F$; and the thesis examples $N_3 : 2X_1 \rightleftarrows X_1 + X_2 \to 2X_2$ and $N_{4.12} : 2X_1 \rightleftarrows X_1 + X_2$.

Theorem 11 (the precise forms fail Thm 1 [6])**.** Read "duality relation" as equality, refinement in either direction or equal number of blocks, on either side of the duality. Then Conjecture 22, the siphon versions of Question 24 and the claim of Remark 23 are false; every entry of Table 10 is a counterexample satisfying (H1)–(H3).

Proof. For T: no doubling pair, so $\mathsf{P}(T)$ and $\mathsf{P}(T^*)$ are discrete; the flux cone and the conservation cone are the ray $(1, 1, 1)$, so $\mathsf{M}(T)$ and $\mathsf{M}(T^*)$ are one block; the only siphon is $\{X_1, X_2, X_3\}$; the internal cycle $\{R_1, R_2, R_3\}$ lies in no precluster. The other

entries are checked the same way (`certify_duality.py`, claims "S3 table" to "S3 N4.12"; the archived script certifies 142 claims). □

The archived report also checks the criteria on all 15 513 networks satisfying (H1)–(H3) in five exhaustive small families §4 [6]. The counterexamples persist in the Petri class: R, T, W, W^*, F are well-formed free-choice with well-formed free-choice duals (Theorem 24).

3.2 Definition 19 is not invariant

Splitting a three-way choice into nested binary choices, or duplicating a reaction (two copies with rates adding up to the old one give the same dynamics), should not change a structural invariant. Definition 19 is not invariant.

Proposition 12 (agents P, Q2)**.** Let $B_k : X \to Y_a$ $(a = 1, \dots, k)$, $Y_a \to Z$, $Z \to X$, a well-formed free-choice network with (H1)–(H3). Then $\mathsf{P}(B_2)$ is one block, while $\mathsf{P}(B_3) = \{\text{all but } Z \to X\}, \{Z \to X\}$; splitting the choice of B_3 through a new species W ($X \to Y_1$, $X \to W$, $W \to Y_2$, $W \to Y_3$) gives one block again. Duplicating the reaction $2X_1 \to X_1 + X_2$ of $N_{4.12}$ merges its two preclusters.

Proof. The T-components of B_k are $\{X \to Y_a, Y_a \to Z, Z \to X\}$, so $n_{X \to Y_a} = n_{Y_a \to Z} = e_a$ and $n_{Z \to X} = e_1 + \cdots + e_k$, which lies in $\operatorname{span}\{e_a, e_b\}$ iff $k = 2$. For $N_{4.12}$ with R_1 duplicated, $n_{R_2} = n_{R_1} + n_{R_1'}$ lies in the span of the new doubling pair. □

3.3 The corrected preclusters

Group the reactions by source complex (m classes, indicator matrix $Q \in \{0,1\}^{r \times m}$). For $k \in \mathbb{R}^r_{>0}$ let $A_k = \Gamma \operatorname{diag}(k) Q$: its column c is $\sum_{t:\, y_t = c} k_t (y_t' - c)$, and the mass-action field is $f(x) = A_k \varphi(x)$ with $\varphi_c(x) = x^c$. This is the complex-to-species rate matrix of Pérez Millán, Dickenstein, Shiu and Conradi [28] restricted to source complexes. Let $C^\circ = \ker \Gamma \cap \mathbb{R}^r_{>0}$, which under (H2) is $\{E\lambda : \lambda > 0\}$. For $V \in C^\circ$, $\mathbf{1} \in \ker A_V$.

Definition 13 (agent Q2)**.** Assume (H2). κ_V $(V \in C^\circ)$ groups source classes c, c' with $u_c = u_{c'}$ for every $u \in \ker A_V$. The *corrected preclusters* are $\widehat{\mathsf{P}}(\mathcal{N}) = \bigwedge_{V \in C^\circ} \kappa_V$ (meet). The *class-span closure* P_{cs} merges a block C into a block B with $|B| \geq 2$ when some $s \in C$ has $n_s \in \operatorname{span}\{n_t : t \in B\}$; the *flat closure* π is the finest partition coarser than the source classes whose blocks B satisfy $[n_s \in \operatorname{span}\{n_t : t \in B\} \Rightarrow s \in B]$.

Lemma 14 (what Conradi–Flockerzi and the thesis compute; agent Q2, reading of V1)**.** Assume (H2). With the clusters of Conradi and Flockerzi Fact 2(b), Alg. 1 [24] read as identities on the pairs (λ, μ) satisfying the doubling relations (their Fact 2(a); equivalently the thesis Definition 4.6), their cluster partition is $\widehat{\mathsf{P}}(\mathcal{N})$. For fixed k the thesis clusters are the classes of source complexes whose coordinate functionals are proportional on $\ker A_k$.

Proof. For $V, W \in C^\circ$, W/V is constant on source classes iff $W = \operatorname{diag}(V) Q \omega$ with $\omega \in \ker A_V \cap \mathbb{R}^m_{>0}$. The doubling relations say exactly that W/V is constant on doubling pairs; the admissible W form a relatively open subset of $\operatorname{diag}(V) Q \ker A_V$, and two linear functionals agree on it iff they agree on $\ker A_V$. □

V1 notes that the domain of the identity in Conradi–Flockerzi is not stated precisely: read over all pairs of $\Lambda(E)^2$ it would give the full-coupling classes, and read over pairs with the full Fredholm constraint it can only be coarser. We use the reading of the lemma.

Proposition 15 (hierarchy; agent Q2)**.** Under (H2), $\mathsf{P} \preceq \mathsf{P}_{\mathrm{cs}} \preceq \pi \preceq \widehat{\mathsf{P}} \preceq \kappa_V$ for every $V \in C^\circ$, and $\pi = \mathsf{P}_{\mathrm{cs}} \vee \mathsf{M}(\mathcal{N}^*)$. All inclusions can be strict: B_3 ($\mathsf{P} \prec \mathsf{P}_{\mathrm{cs}}$); R ($\mathsf{P}_{\mathrm{cs}} \prec \pi$); the complete digraph K_3 on X, Y, Z ($\pi =$ three source classes, $\widehat{\mathsf{P}}$ one block); $XY_4 \colon X + Y \to 2X,\ X + Y \to 2Y,\ X + 2Y \to 2X + Y,\ X + 2Y \to 3Y$ ($\widehat{\mathsf{P}} = \pi = \{R_1, R_2\}, \{R_3, R_4\}$ while κ_V is one block for generic V and two blocks at $V = (1, 1, 2, 2)$).

Proof. $\pi \preceq \kappa_V$ is the ratio argument of Conradi–Flockerzi made class-wise: if $w = Q\omega$ ($\omega \in \ker A_V$) is constant ρ on a set B, then $\operatorname{diag}(V)w - \rho V \in \ker \Gamma$ vanishes on B, hence at every s with $n_s \in \operatorname{span}\{n_t : t \in B\}$. The closure of a singleton is its fully coupled class, which gives $\pi = \mathsf{P}_{\mathrm{cs}} \vee \mathsf{M}(\mathcal{N}^*)$. The witnesses are certified exactly; for K_3 and for B_2 with $Z \to X$ duplicated, "$\widehat{\mathsf{P}}$ is one block" is certified by an $(m-1)$-minor of $A_{E\lambda}$ whose coefficients as a polynomial in λ all have one sign. □

Theorem 16 (invariance; agent Q2)**.** $\widehat{\mathsf{P}}$ and the family $(\kappa_V)_V$ are invariant under relabelling, duplication of a reaction, and splitting a choice (or subdividing a reaction) through a new intermediate species: $\widehat{\mathsf{P}}(\mathcal{N}') = \varphi^{-1}(\widehat{\mathsf{P}}(\mathcal{N}))$ for the natural map φ of reactions. P, P_{cs} and π are not invariant. Any rule X with $X(\mathcal{N}) \succeq \mathsf{P}(\mathcal{N})$ that is invariant under duplication satisfies $X(\mathcal{N}) \succeq \mathsf{P}(\mathcal{N}) \vee \mathsf{M}(\mathcal{N}^*)$.

Proof. Duplication: $A'_{V'} = A_V$ with V the image of V', and $V' \mapsto V$ maps $C^\circ(\mathcal{N}')$ onto $C^\circ(\mathcal{N})$. Splitting $A \subseteq c$ behind W: the new row W forces $\omega'_{c'} = \omega'_A$ and the two new columns add up to the old column c, so $\ker A'_{V'} \cong \ker A_V$. Non-invariance: Proposition 12; B_2 versus B_2 with $Z \to X$ duplicated ($\mathsf{P}_{\mathrm{cs}} = \pi$ one block, then two blocks). Forcing: if s, t are fully coupled, duplicate t; then $n'_s \in \operatorname{span}\{n'_t, n'_{t'}\}$ and s is adjoined to the doubling pair $\{t, t'\}$. □

So any invariant repair of Definition 19 contains the full-coupling classes. $\widehat{\mathsf{P}}$ is such a repair; V1 points out that it is not shown to be the finest one, and we do not call it canonical.

Theorem 17 (structure of $\widehat{\mathsf{P}}$; agents Q2, E2)**.** Assume (H2).

(1) If $\mathcal{N}$ is EC, connected and conservative, $\ker A_V = \mathbb{R}\mathbf{1}$ for every $V \in C^\circ$, so $\widehat{\mathsf{P}}(\mathcal{N})$ is one block.

(2) If $\mathcal{N}$ is weakly reversible with deficiency zero, $\ker A_V = \operatorname{span}\{\mathbf{1}_L\}$ over the linkage classes L, so $\widehat{\mathsf{P}}(\mathcal{N})$ is the linkage-class partition.

(3) If every linkage class carries a positive flux of its own (e.g. $\mathcal{N}$ weakly reversible), $\widehat{\mathsf{P}}(\mathcal{N})$ refines the linkage classes, whatever the deficiency.

Proof. (1) By Lemma 7 and Lemma 27 below, a nonzero nonnegative vector of $\ker A_V$ is positive and spans the kernel. (2) $\Gamma = YI$ with complex matrix Y and incidence I;

$A_V = Y L_V$ with $L_V = I\,\mathrm{diag}(V)Q$; deficiency zero gives $\ker A_V = \ker L_V$, the Laplacian kernel of a weakly reversible graph, and $\mathbf{1} \in \ker A_V$ fixes the basis. (3) With $V = (V_L)_L$, $\mathbf{1}_L \in \ker A_V$ separates L from the other classes. □

Proposition 18 (toric steady states; agent Q2)**.** Let $\psi \in \ker A_k \cap \mathbb{R}^m_{>0}$. Then $\dim\ker A_k \le \#\kappa(k)$, with equality iff $\ker A_k$ has a basis with disjoint supports (Condition 3.1 of [28]); then the positive steady states are $\{x > 0 : x^{y_c - y_{c'}} = \psi_c/\psi_{c'} \text{ for } c \sim c'\}$, a toric set. Moreover $\mathrm{rank}\, A_V \ge m - \#\widehat{\mathsf{P}}(\mathcal{N})$, and equality for every $V \in C^\circ$ means toric steady states for every rate vector with the rate-independent partition $\widehat{\mathsf{P}}(\mathcal{N})$. *Remark 23 for* $\widehat{\mathsf{P}}$: every internal cycle lies in a block of $\widehat{\mathsf{P}}(\mathcal{N})$ iff $\ker\Gamma = \bigoplus_{B\in\widehat{\mathsf{P}}}(\ker\Gamma \cap \mathbb{R}^B)$ (the isolation property of Conradi and Flockerzi), and this implies the equality.

Proof. In each class the coordinate functionals are proportional with factors fixed by ψ, so $u \mapsto (u_{c_J})_J$ is injective on the kernel, with equality iff each $\psi|_J$ lies in it. Under isolation, the flux $\mathrm{diag}(V)Q\omega$ splits into its parts on the blocks, so $\omega|_B \in \ker A_V$ for each block and $\dim\ker A_V \ge \#\widehat{\mathsf{P}} \ge \#\kappa_V \ge \dim\ker A_V$. □

"Isolation implies toric" is Iosif's thesis Theorem 4.17 in a rate-dependent form; the rate-free form with $\widehat{\mathsf{P}}$ was not found in the literature as a statement (Section 9.3). XY_4 and $Z_4\colon 2X_2 \to 2X_1,\ 2X_1 \to 2X_2,\ X_1 + X_2 \to 2X_1,\ X_1 + X_2 \to 2X_2$ fail Remark 23 for $\widehat{\mathsf{P}}$; the 3-cycle T now satisfies it.

Theorem 19 (Conjecture 22 for $\widehat{\mathsf{P}}$; agent Q2, with V1's wording)**. (1)** If $\mathcal{N}$ is consistent, then $\mathsf{M}(\mathcal{N}^*) \preceq \widehat{\mathsf{P}}(\mathcal{N})$; and $\widehat{\mathsf{P}}(\mathcal{N}) = \mathsf{M}(\mathcal{N}^*)$ iff $\widehat{\mathsf{P}}(\mathcal{N}) \preceq \mathsf{M}(\mathcal{N}^*)$ iff both have the same number of blocks iff every doubling pair is fully coupled. The same holds with π or any κ_V in place of $\widehat{\mathsf{P}}$.

(2) For every conservative network, $\mathsf{M}(\mathcal{N}) \preceq \widehat{\mathsf{P}}(\mathcal{N}^*)$, with equality iff every twin pair is conservation-coupled. If moreover (H3) holds and every product complex is a source complex (in particular for every weakly reversible network satisfying (H1)–(H3)), $\widehat{\mathsf{P}}(\mathcal{N}^*) = \mathsf{M}(\mathcal{N})$.

(3) If $\mathcal{N}$ is weakly reversible without loops, $\widehat{\mathsf{P}}(\mathcal{N}) = \mathsf{M}(\mathcal{N}^*)$ iff every linkage class is a directed cycle.

Proof. $\mathsf{M}(\mathcal{N}^*) \preceq \pi \preceq \widehat{\mathsf{P}} \preceq \kappa_V$ (Proposition 15). If every doubling pair is fully coupled, $\mathrm{diag}(V)^{-1}\ker\Gamma \subseteq Q(\mathbb{R}^m)$ equals $Q\ker A_V$, so $\kappa_V = \pi = \widehat{\mathsf{P}} = \mathsf{M}(\mathcal{N}^*)$. Refinement and equal cardinality give equality. (2) is (1) for $\mathcal{N}^*$ with Theorem 9; (3) uses the argument of Theorem 9 for weakly reversible networks. □

The refinement is built in ($\pi = \mathsf{P}_{\mathrm{cs}} \vee \mathsf{M}(\mathcal{N}^*)$), and the surviving counterexamples are those of the criterion: U, F, W^* on the species side, W, N_3, F on the reaction side. The archived Corollary "$\mathsf{P}(\mathcal{N}) \ne \mathsf{M}(\mathcal{N}^*)$ at deficiency zero" is an artefact of Definition 19.

3.4 Siphons: transversality, not refinement

Theorem 20 (transversality; agent P)**.** Let R be a minimal siphon of $\mathcal{N}$ and $i \neq j$ species with the same consuming reactions ($i^\bullet = j^\bullet$; twins in particular). Then R does not contain both. Dually, a minimal T-trap contains no two reactions with the same reactant set. In an EC network every minimal siphon meets every cluster in at most one species.

Proof. If $i, j \in R$, then $R\backslash\{i\}$ is a siphon: a reaction producing a species of $R\backslash\{i\}$ consumes a species of R; if that species is i, the reaction lies in $i^\bullet = j^\bullet$ and consumes j. The dual statement is the same argument for $\mathcal{N}^*$. □

For free-choice nets this is Desel and Esparza Thm 4.30(a) [20], with the same proof; the weighted statement is a mild generalisation.

Theorem 21 (when preclusters lie in minimal siphons; agent P, corrected by V1)**.** Every precluster of $\mathcal{N}^*$ lies in a minimal siphon of $\mathcal{N}$ iff $\mathcal{N}$ has no two distinct species with equal rows of Y_e (zero rows included) and every species lies in a minimal siphon. Dually, every precluster of $\mathcal{N}$ lies in a minimal T-trap iff $\mathcal{N}$ has no doubling pair and every reaction lies in a minimal T-trap.

Proof. A precluster of $\mathcal{N}^*$ with two elements contains a twin pair, and then lies in no minimal siphon by Theorem 20. Without twins $\mathsf{P}(\mathcal{N}^*)$ is discrete. □

V1's correction concerns the definition of twins: with twins restricted to nonzero rows the statement fails for $X_1 \rightleftarrows X_2$ with two further species X_3, X_4 that occur in no reaction ($\mathsf{P}(\mathcal{N}^*) = \{X_1\}, \{X_2\}, \{X_3, X_4\}$, minimal siphons $\{X_3\}, \{X_4\}, \{X_1, X_2\}$); (H3) excludes this case. So where preclusters are nontrivial they are transversal to the minimal siphons, which explains the failures of Q3 in Table 10. Two weaker relations survive.

Proposition 22 (MIPS refine preclusters away from singletons; agent P)**.** Under (H1), every precluster of $\mathcal{N}^*$ with at least two elements is a union of blocks of $\mathsf{M}(\mathcal{N})$; under (H2), every precluster of $\mathcal{N}$ with at least two elements is a union of fully coupled classes.

Proof. If s is in a twin pair $\{s, j\}$, or adjoined to $\{i, j\}$, and $w_{s'} = \mu w_s$, then $w_{s'}$ lies in the same span, so s' is adjoined too. □

Proposition 23 (MIPS and siphons; agent P)**.** Under (H1), if the minimal siphons are exactly the supports of the extreme conservation laws, then every minimal siphon is a union of blocks of $\mathsf{M}(\mathcal{N})$ and every block lies in a minimal siphon. This holds when $\mathcal{N}$ has no critical siphon, e.g. under the cluster rank equation (Theorem 31). D refutes all three statements ($\mathsf{M}(D)$ is one block, the minimal siphons are $\{X_1, X_2\}, \{X_1, X_3\}$); $N_{4.12}$ refutes "every support of an extreme law is a union of minimal siphons" (law $X_1 + X_2$, only minimal siphon $\{X_1\}$).

Proof. Conservation-coupled species ($w_i = \mu w_j$) lie in the same supports of extreme laws, and by (H1) every species lies in one. If every siphon contains the support of a nonnegative law, a minimal siphon equals such a support, which is then the support of ⤵

an extreme law, and conversely. The examples are certified (`certify_duality.py`). □

For $\widehat{\mathsf{P}}$ nothing new appears: under (H1), every block of $\widehat{\mathsf{P}}(\mathcal{N}^*)$ lies in a minimal siphon iff $\mathcal{N}$ has no twin pair and every block of $\mathsf{M}(\mathcal{N})$ lies in a minimal siphon (Theorems 20 and 19). In the free-choice class the preclusters are determined by the minimal siphons, and this is the most precise positive form of Question 24 we found.

Theorem 24 (well-formed free-choice networks; agent P, from Desel–Esparza)**.** Let $\mathcal{N}$ be well-formed free-choice (free-choice, connected, (H1), (H2), $s = q - 1$). Then $\mathcal{N}^*$ is well-formed free-choice (Hack's duality theorem); the extreme conservation laws are the 0/1 indicators of the minimal siphons (the S-components), and the extreme fluxes those of the minimal T-traps; $\mathsf{M}(\mathcal{N})$ is the partition into atoms of the family of minimal siphons; neither side has a critical siphon; no twin pair is conservation-coupled and no doubling pair is fully coupled; and $\mathsf{P}(\mathcal{N}^*)$ is the component partition of the graph with edges $\{i, j\}$ (twins) and $\{i, s\}, \{j, s\}$ whenever $\Sigma(s) \in \{\Sigma(i), \Sigma(j), \Sigma(i) \cup \Sigma(j)\}$, $\Sigma(i)$ being the set of minimal siphons containing i. If $n, r \geq 2$, $\mathsf{P}(\mathcal{N}^*) \neq \mathsf{M}(\mathcal{N})$ and $\mathsf{P}(\mathcal{N}) \neq \mathsf{M}(\mathcal{N}^*)$.

Proof (sketch)**.** Rank Theorem 6.14, Proposition 5.4 and Theorem 6.21 of [20]; the profile rule follows because the columns of the law matrix are the indicators of $\Sigma(i)$ and twins have disjoint Σ by Theorem 20. Full proof in the notes of agent P, re-derived by V1. □

Proposition 25 (no siphon description outside free choice; agents P, Q3)**.** The networks $M_3 \colon X_4 \to 2X_3,\ X_4 \to X_2 + X_3,\ 2X_3 \to 2X_2,\ X_2 + X_3 \to 2X_1,\ 2X_2 \to X_4,\ 2X_1 \to 2X_3$ and $M_4 \colon X_4 \to X_2 + X_3,\ 2X_3 \to X_4,\ 2X_3 \to 2X_2,\ X_2 + X_3 \to 2X_1,\ 2X_2 \to X_4,\ 2X_1 \to 2X_3$ are weakly reversible and pure, satisfy (H1)–(H3) on both sides, and have the rows and the columns of Γ pairwise linearly independent. Exactly eight relabellings of the reactions match their doubling classes and minimal T-traps, and the preclusters differ under all eight. By duality, M_3^* and M_4^* have the same twins and minimal siphons but different P of their duals.

Proof. Checked by computation in exact arithmetic (`certify_duality.py`, claims "S3 K8"); agents P and Q3, re-checked by V1 and V2. □

Answer to Q1. Preclusters and siphons are related by transversality (Theorems 20 and 21), not by refinement; in the Petri class the preclusters of $\mathcal{N}^*$ are a function of the twins and the minimal siphons of $\mathcal{N}$ (Theorem 24), and outside it they are not (Proposition 25). Maximal invariant polyhedral supports and corrected preclusters correspond exactly in one direction, and with equality on the species side of the GAC setting (Theorem 19). Agent E's Proposition E.3 shows that a Galois connection between these lattices always exists and therefore carries no information; every "union" and "singletons removed" reading fails already on two species (agent E, checked by computation; `gac-duality/computation/notes.md` §3, recounted by V1).

4 What the duality transfers

4.1 Standing lemmas

For a network with Petri clusters $c = (S_c, T_c)$, $c = 1, \dots, q$, and $k \in \mathbb{R}^r_{>0}$, let $\bar{A}_k$ be the $n \times q$ matrix with columns $\Gamma(k \odot \mathbf{1}_{T_c})$. Since a species of S_c is a reactant of no reaction outside T_c,

$$\bar{A}_k[p, d] = \textstyle\sum_{t \in T_d} k_t (y'_t)_p \geq 0 \quad (p \in S_c,\ d \neq c), \qquad \bar{A}_k \mathbf{1} = \Gamma k. \tag{26}$$

The entry is positive iff some reaction of T_d produces p, i.e. iff $d \to c$ in the cluster graph. For EC networks $\bar{A}_k = A_k$ (Section 3.3), and the mass-action field is $A_k \varphi(x)$ with $\varphi_c(x) = x^{y_c}$. We write $D = \operatorname{span}\{y_c - y_{c'}\}$ for the span of the differences of source complexes and $L = S^\perp$ for the conservation laws.

Lemma 27 (agents P, Q3)**.** Let every species be consumed and the Petri net be strongly connected. If $u \geq 0$, $u \neq 0$ and $\bar{A}_k u = 0$, then $u > 0$ and $\ker \bar{A}_k = \mathbb{R}u$.

Proof. Row $p \in S_c$ reads $(\sum_{t \in T_c} k_t (y_t)_p) u_c = \sum_d (\sum_{t \in T_d} k_t (y'_t)_p) u_d$ with a positive coefficient on the left, so $u_d > 0$ and $d \to c$ give $u_c > 0$: the support is closed in the strongly connected cluster graph (Lemma 7). If $w \in \ker \bar{A}_k$ is not proportional to u, $u - \varepsilon w$ for the largest admissible ε is a nonzero nonnegative kernel vector with a zero entry. □

Lemma 28 (Perron–Frobenius for Metzler matrices)**.** Let G be an irreducible Metzler matrix with $Gw = 0$ for some $w > 0$ (or $\xi^{\mathsf{T}} G = 0$ for some $\xi > 0$). Then 0 is a simple eigenvalue and every other eigenvalue has negative real part. If G is Metzler, possibly reducible, with $G\mathbf{1} = 0$, some nonzero $\xi \geq 0$ satisfies $\xi^{\mathsf{T}} G = 0$.

Proof. For $c > 0$ large, $G + cI$ is nonnegative, irreducible and has a positive diagonal, hence is primitive. A positive null vector of G (or of G^{T}) is a positive eigenvector of $G + cI$ (or of its transpose) for the eigenvalue c, so by the Perron–Frobenius theorem c is the spectral radius and a simple eigenvalue, and every other eigenvalue $\lambda + c$ satisfies $|\lambda + c| < c$, i.e. $\operatorname{Re} \lambda < 0$. In the reducible case $G + cI$ is nonnegative with all row sums c, so its spectral radius is c, and the Perron–Frobenius theorem for nonnegative matrices gives a nonzero $\xi \geq 0$ with $\xi^{\mathsf{T}}(G + cI) = c\,\xi^{\mathsf{T}}$. □

Lemma 29 (routing; agents P, Q3)**.** Let $\mathcal{N}$ be connected with (H1), (H2) and $s = q - 1$. For every $k \in \mathbb{R}^r_{>0}$, $\ker \bar{A}_k = \mathbb{R}u(k)$ with $u(k) > 0$, and $\operatorname{im} \bar{A}_k = S$.

Proof. $\operatorname{rank} \bar{A}_k \leq s < q$, so the kernel is never zero. The set of k whose kernel contains a nonzero $u \geq 0$ is closed (normalise and take limits), contains a positive flux v (with $u = \mathbf{1}$, by (26)), and is open: there the rank is $q - 1$ by Lemma 27, it stays $q - 1$ nearby, and the kernel line moves continuously through a positive vector. So it is all of $\mathbb{R}^r_{>0}$. Then $\operatorname{rank} \bar{A}_k = q - 1 = s$ and $\operatorname{im} \bar{A}_k \subseteq S$. □

4.2 The cluster rank equation: no critical siphon on either side

Proposition 30 (agents P, Q3)**.** If $\mathcal{N}$ is connected with (H1) and (H2), then $s \geq q - 1$.

Proof. Lemma 27 with $k = v > 0$ in $\ker \Gamma$ and $u = \mathbf{1}$: $q - 1 = \operatorname{rank} \bar{A}_v \leq s$. □

Theorem 31 (cluster rank equation; agent Q3)**.** Let $\mathcal{N}$ be connected with (H1), (H2) and $s = q - 1$. Then every siphon of $\mathcal{N}$ and every siphon of $\mathcal{N}^*$ contains the support of a nonzero nonnegative conservation law. Hence neither $\mathcal{N}$ nor $\mathcal{N}^*$ has a critical siphon, the minimal siphons of $\mathcal{N}$ are the supports of its extreme conservation laws, the minimal T-traps of $\mathcal{N}$ are its internal cycles, and $\mathcal{N}$ and $\mathcal{N}^*$ are persistent for every rate vector.

Proof. Let R be a siphon, C_R the clusters whose species meet R, $v > 0$ in $\ker \Gamma$ and $B = \bar{A}_v[R, C_R]$. By (26), $B[p, d] \geq 0$ for $p \in R \cap S_c$, $d \neq c$. For $p \in R$ and $d \notin C_R$, $\bar{A}_v[p, d] = 0$: a reaction producing p consumes a species of R, so its cluster is in C_R. Hence $B\mathbf{1} = (\Gamma v)_R = 0$. If no $z \geq 0$, $z \neq 0$ had $z^{\mathsf{T}} B = 0$, Gordan's alternative would give w with $Bw > 0$; at a maximiser c^* of w and $p \in R \cap S_{c^*}$, $(Bw)_p = \sum_d B[p, d](w_d - w_{c^*}) \leq 0$, a contradiction. Extended by zero, $z^{\mathsf{T}} \bar{A}_v = 0$, and $\operatorname{im} \bar{A}_v = S$ by Lemma 29, so $z^{\mathsf{T}} \Gamma = 0$. The dual $\mathcal{N}^*$ is connected, has rank s, the same clusters, and (H1), (H2) exchanged (Table 6), so it satisfies the same hypotheses. The support of a nonzero conservation law $z \geq 0$ is a siphon, which gives the statement on minimal siphons; for T-traps apply it to $\mathcal{N}^*$. Persistence for every rate vector is Theorem 2 of Angeli, De Leenheer and Sontag [8] (a conservative network whose siphons all contain the support of a nonnegative conservation law is persistent, for mass action and more general kinetics). □

Corollary 32 (well-formed equal-conflict networks)**.** If $\mathcal{N}$ is WF-EC (no purity, no balanced joins), $\mathcal{N}$ and $\mathcal{N}^*$ have no critical siphon, the minimal siphons of $\mathcal{N}$ are the internal cycles of $\mathcal{N}^*$ and conversely, and both are persistent for every rate vector.

Proof. Theorem 31 with $q = m$ (Section 2); the internal cycles of $\mathcal{N}^*$ are the supports of the extreme laws of $\mathcal{N}$ (Lemma 2). □

Credits. For ordinary nets the siphon statement of Theorem 31 is essentially known: Desel and Esparza Thms 10.16–10.17 [20] (Theorem 10.16 is due to Desel [29]) show that an ordinary net with positive S- and T-invariants and the rank equation is well-formed and that a marking is live and bounded iff it marks every proper siphon; their proof shows that a marking marking every S-component is live, and two more lines give that every siphon contains an S-component (observed by L2; V1 read the proof of 10.17, V2 did not re-derive that step). For free-choice nets it is Desel–Esparza Proposition 5.4. Teruel and Silva give the rank theorem for weighted EC nets but write that siphons have no satisfactory weighted theory pp. 272, 296 [21]. For ordinary nets the statement for the reverse dual follows in the same way, since the hypotheses of Theorem 10.16 are invariant under it (Desel–Esparza Prop. 6.20). The weighted statement (for $\mathcal{N}$ and for $\mathcal{N}^*$), its persistence reading and Proposition 30 outside free choice were not found (Section 9.3). Agent P first proved Corollary 32 for pure WF-EC networks through Teruel–Silva's Theorems 28, 15 and Corollary 13, and its dual half by a limit of rate constants; agent Q3's proof above needs neither purity nor Teruel–Silva.

4.3 Well-formed equal-conflict networks

Theorem 33 (toric steady states; agent P)**.** Let $\mathcal{N}$ be EC, connected, with (H1), (H2) and m source complexes.

(1) If $s = m - 1$, then for every k, $\ker A_k = \mathbb{R}u(k)$ with $u(k) > 0$, and the positive steady states are $\{x > 0 : x^{y_c} = \lambda u_c(k)\ \forall c,\ \lambda > 0\}$, a nonempty set whose logarithm is an affine space of dimension $n - s$.

(2) Conversely, if $\mathcal{N}$ has a positive steady state for every k, then $s = m - 1$.

Proof. (1) is Lemma 29 with $q = m$; the sources have disjoint supports, so they are linearly independent and the $m - 1$ log-linear equations $(y_c - y_{c_0}) \cdot \log x = \log(u_c/u_{c_0})$ are independent and solvable. (2) If $s \geq m$, the rate vectors with a positive steady state are $k_t = v_t/u_{c(t)}$ with $v \in \ker \Gamma \cap \mathbb{R}^r_{>0}$ and $u \in \mathbb{R}^m_{>0}$, the image of a set of dimension $(r - s) + m - 1 < r$; it has measure zero; with Proposition 30, $s = m - 1$. □

Credits. Once $\ker A_k$ is a positive line, the toric conclusion is a special case of Pérez Millán, Dickenstein, Shiu and Conradi Thm 3.8, Cor. 3.12 [28]. The flux half for ordinary free-choice nets with rational routing rates is Campos, Chiola and Silva Lemma 5.1, Thms 5.3–5.4 [30]; for continuous equal-conflict nets under infinite-server (min) semantics the routing relation is in Recalde, Júlvez and Silva [31] and Júlvez, Recalde and Silva eq. (18), Def. 7 [32]. None of these has a uniqueness or stability result under mass action. Not found in the literature (Section 9.3): weighted EC under mass action, all real rates, the converse (2), and the dual statements below. Johnston's translated networks [33] do not cover WF-EC in general (V2): $A \to C,\ 2C \to 2A$ is WF-EC, but $\ker \Gamma = \mathbb{R}(2, 1)$ is not the cycle space of a graph, so it has no translation of deficiency zero.

Corollary 34 (self-duality with balanced joins; agent P)**.** If $\mathcal{N}$ is EC with balanced joins (e.g. free-choice), connected, with (H1), (H2), then $\mathcal{N}$ has positive steady states for all rate vectors iff $\mathcal{N}^*$ does iff $s = q - 1$; both steady-state sets are then toric, of dimensions $n - s$ and $r - s$.

Proof. $\mathcal{N}^*$ is EC with balanced joins, connected, of rank s, with the same clusters and (H1), (H2) exchanged (Table 6); apply Theorem 33 to $\mathcal{N}$ and to $\mathcal{N}^*$. □

Theorem 35 (uniqueness and local stability; agent Q3)**.** Let $\mathcal{N}$ be WF-EC and $k \in \mathbb{R}^r_{>0}$.

(1) Every positive compatibility class contains exactly one positive steady state.

(2) At every positive point x, steady or not, $\det(tI - J(x)) = t^{n-s}h(t)$ with h Hurwitz: $J(x)$ restricted to S has all eigenvalues in the open left half-plane.

(3) $\sigma(S) \cap \sigma(D^\perp) = \{0\}$ (σ = sign vectors).

Proof. (1), uniqueness. Let x, x' be positive steady states with $x - x' \in S$. By Theorem 33, $\delta := y_c \cdot (\log x - \log x')$ does not depend on c. Write $x - x' = A_v w$ ($v > 0$ ⤵

in $\ker\Gamma$, Lemma 29). At a maximiser c^* of w and $p \in S_{c^*}$, $(x-x')_p = \sum_d A_v[p,d](w_d - w_{c^*}) \le 0$ by (26), so $\delta \le 0$; at a minimiser $\delta \ge 0$. Hence $\delta = 0$, $x = x'$ on the maximising clusters, the set of maximisers is closed under predecessors, w is constant and $x = x'$. Existence: Brouwer's theorem on the compact class gives an equilibrium; its zero set would be a siphon, which contains the support of a conservation law (Corollary 32), impossible. (3) is the same argument with $x - x'$ and $\log x - \log x'$ replaced by vectors of S and $D^\perp$ with equal signs. (2) $J = A_k D\varphi(x)$ with $D\varphi(x)[c,p] = (y_c)_p \varphi_c(x)/x_p$. The $m \times m$ matrix $G = D\varphi(x)A_k$ is Metzler and irreducible by (26) and Lemma 7, and $Gu(k) = 0$ for every x. Lemma 28 and $\det(tI_n - A_k D\varphi) = t^{n-m}\det(tI_m - D\varphi A_k)$ give the claim, since J maps into S and induces 0 on $\mathbb{R}^n/S$. □

For mass action, (3) is equivalent to "at most one steady state per class for every k" by the sign-vector criterion of Müller and Regensburger Prop. 3.1 of the arXiv version [34], since $u(k)$ ranges over all of $\mathbb{R}^m_{>0}$. The Metzler/null-vector argument of (2) is the Perron–Frobenius device behind the spectral comparison of ΓG and $G\Gamma$ used by Jamshidi and Palsson [19] and by Blanchini and Giordano Thm 1 [16].

Remark 36 (cluster kinetics; correction of V2)**.** The proof of (1) works for rates $k_t\varphi_c(x)$ with φ_c a C^1 function of the species of S_c, strictly increasing in each and vanishing iff one vanishes. Part (2), and Theorem 37 when φ_c is moreover analytic, need in addition $\partial\varphi_c/\partial x_p > 0$ on the open orthant for $p \in S_c$; strict monotonicity alone does not suffice ($\varphi(x) = (x-1)^3 + 1$ has a zero derivative at $x = 1$). ⌟

Theorem 37 (global stability; agent Q3)**.** Let $\mathcal{N}$ be WF-EC, $k \in \mathbb{R}^r_{>0}$ and $u = u(k)$. Along every positive trajectory, $M(t) = \max_c x^{y_c}/u_c$ is nonincreasing and $\mu(t) = \min_c x^{y_c}/u_c$ is nondecreasing, and $x(t)$ converges to the steady state of its class.

Proof. $\psi = \varphi(x)$ satisfies $\dot\psi = G(x)\psi$ with G Metzler and $Gu = 0$; with $h_c = \psi_c/u_c$, $\dot h_c = \sum_{d\neq c} G[c,d](u_d/u_c)(h_d - h_c)$, a consensus system with nonnegative state-dependent weights. So the upper Dini derivative of M is ≤ 0 (ties are harmless) and that of μ is ≥ 0. Then $\psi_c(t) \ge \mu(0)u_c$, x is bounded above by (H1) and every species lies in some S_c, so the trajectory stays in a compact subset of the open orthant. $W = \log M - \log\mu$ vanishes exactly at steady states and is constant, say W^*, on the ω-limit set. If $W^* > 0$, along an orbit in it M and μ are constant; by analyticity some $h_c \equiv M$, and the identity for $\dot h_c$ forces every predecessor of c to share this value, so all do and $W = 0$, a contradiction. So the ω-limit set consists of steady states of the class, and Theorem 35 ends the proof. □

The max/min monotonicity is the classical argument of nonlinear compartmental and consensus systems [35] [36] (not opened here; cited after L2), and the functions are of the piecewise-linear-in-rates type of Al-Radhawi and Angeli [37], but depend on k. WF-EC networks are in general neither weakly reversible nor complex balanced (W^*, F; the networks B_k, by contrast, are weakly reversible with deficiency zero), so Theorem 37 is a global-attractor statement outside the complex-balanced setting. For weakly reversible EC networks without the zero complex all complexes have disjoint supports and the deficiency is zero (Mairesse and Nguyen Lemma 3.2, Cor. 3.2 [38] for ordinary nets), so the GAC was already classical there.

4.4 The dual of a well-formed equal-conflict network

Let $\mathcal{N}$ be WF-EC. The reaction ρ_p of $\mathcal{N}^*$ (one per species $p \in S_c$) consumes $a_p = (y_c)_p$ of every K_t, $t \in T_c$; with $\pi_c = \prod_{t\in T_c} K_t$ its rate is $\omega_p = \kappa_p \pi_c^{a_p}$, and a positive K is a steady state iff $\omega \in L$. $\mathcal{N}^*$ is EC iff $\mathcal{N}$ has balanced joins.

Theorem 38 (existence for all rates; agent Q3)**.** If $\mathcal{N}$ is WF-EC, then for every $\kappa \in \mathbb{R}^n_{>0}$, $\mathcal{N}^*$ has a positive steady state; more precisely, for every $M > 0$ some $\pi > 0$ with $\sum_c \pi_c = M$ has $\omega(\pi) \in L$, and every $K > 0$ with $\prod_{t\in T_c} K_t = \pi_c$ is a steady state.

Proof. By Lemma 29, $\omega \in L$ iff $H_c(\pi) := \omega(\pi) \cdot \bar{A}_v[\cdot, c] = 0$ for all c. Then $\sum_c H_c \equiv 0$, H is cooperative by (26), $H_c \geq 0$ when $\pi_c = 0$ and $H_c \geq -C\pi_c$ on the simplex $\{\pi \geq 0, \sum \pi_c = M\}$, so $\pi \mapsto \pi + \varepsilon H(\pi)$ maps the simplex into itself and Brouwer gives a zero; its zero set is closed under successors in the cluster graph, hence empty. □

Theorem 39 (uniqueness and stability of the dual; agent Q3)**.** If $\mathcal{N}$ is WF-EC and $\kappa > 0$: (a) at every positive point of $\mathcal{N}^*$ the Jacobian restricted to $S^* = \operatorname{im} \Gamma^{\mathsf{T}}$ is Hurwitz; (b) every positive class of $\mathcal{N}^*$ contains exactly one positive steady state, exponentially stable in its class.

Proof. (a) $J^* = P^*Q^*$, and $G^* = Q^*P^*$ is irreducible Metzler, with

$$G^*[c,d] = \sum_{t\in T_c} K_t^{-1} \sum_{p\in S_d} \Gamma_{pt} a_p \omega_p, \qquad \xi^{\mathsf{T}} G^* = (A_{1/K}\xi)^{\mathsf{T}} \operatorname{diag}(a\omega) E_S,$$

so $\xi = u(1/K) > 0$ from Theorem 33 for $\mathcal{N}$ with rates $1/K$ is a positive left null vector, and Lemma 28 applies with the multiplicity count of Theorem 35. (b) The classes are compact polytopes ($\mathcal{N}$ consistent) without boundary steady states (Corollary 32); every zero has index $(-1)^s$ by (a), and the degree of the field on the class is $(-1)^s$ (homotopy to $\hat{K} - K$ for an interior point $\hat{K}$; cf. Thm 1 [39]), so there is exactly one. □

Proposition 40 (agent Q3)**.** If $\mathcal{N}$ is WF-EC and $s \leq 2$, $\mathcal{N}^*$ is globally asymptotically stable in every positive class.

Proof. $s = 1$: a scalar equation on a segment with one hyperbolic interior zero, persistent by Corollary 32. $s = 2$: the divergence on the class is the trace of $J^*|_{S^*}$, negative everywhere by Theorem 39; Bendixson–Dulac excludes periodic orbits and cycles of orbits in the simply connected interior, ω-limit sets avoid the boundary, and Poincaré–Bendixson with a unique sink concludes. □

Proposition 41 (what does not hold for the dual; agent Q3)**.** Let $\mathcal{N}$ be EC, connected, with (H1), (H2).

(1) The converse of Theorem 38 fails: $N_0 \colon 2A + B \to 3A$, $2A + B \to A + 2B$ (EC, unbalanced, (H1)–(H3) and connected on both sides, $s = m = 1$) has positive steady states only for $k_1 = k_2$ ($\dot{a} = (k_1 - k_2)a^2 b$), while $N_0^* \colon 2K_1 + 2K_2 \to 3K_1 + K_2$, $K_1 + K_2 \to 2K_2$ has the steady state $(1, \kappa_B/\kappa_A)$ for every κ. Also $N_1 \colon 2A + B \to C$, $C \to 3A$, $C \to 3B$ ($\ker \Gamma = \mathbb{R}(3,2,1)$) and $N_1^* \colon 2K_1 \to 3K_2$, $K_1 \to 3K_3$, $K_2 + K_3 \to K_1$ with the steady state

$(\kappa_B/\kappa_A, 3\kappa_B^2/(\kappa_A\kappa_C), 1)$.

(2) If $s \geq m+1$, or $s = m$ with balanced joins, $\mathcal{N}^*$ is not absolutely consistent. If $s = m$ with unbalanced joins, both cases occur: N_0, N_1 have absolutely consistent duals, and $NT\colon X_1+X_2 \to 2X_2,\ X_1+X_2 \to 2X_3+X_4,\ 2X_3+X_4 \to 2X_1$ does not (every conservation law has $z_1 = z_2$, while the dual reactions of X_1 and X_2 share the source $K_1 + K_2$).

(3) With unbalanced joins the steady states of $\mathcal{N}^*$ need not be toric: for $N_2\colon 2A + B \rightleftarrows C$ (WF-EC), $N_2^*\colon 2K_1 \to 2K_2,\ K_1 \to K_2,\ K_2 \to K_1$ has the steady states $K_2 = (2\kappa_1 K_1^2 + \kappa_2 K_1)/\kappa_3$, and $d^2 \log K_2/d(\log K_1)^2 = 2\kappa_1\kappa_2 K_1/(2\kappa_1 K_1 + \kappa_2)^2 \neq 0$.

Proof. (2) The admissible κ are $\kappa_p = w_p \pi_{c(p)}^{-a_p}$ with $w \in L \cap \mathbb{R}^n_{>0}$, $\pi \in \mathbb{R}^m_{>0}$: an image of dimension $(n-s)+m$, which is $< n$ if $s \geq m+1$, and $\leq n-1$ with balanced joins since $(w, \pi) \mapsto (tw, t^{1/a}\pi)$ fixes κ. The examples are certified exactly. □

4.5 The self-dual class: proportional conflict

Theorem 42 (agent Q3). Let $\mathcal{N}$ be PC, connected, with (H1), (H2) and $s = q-1$. Then $\mathcal{N}^*$ has the same properties, and for both $\mathcal{N}$ and $\mathcal{N}^*$ and every rate vector: (a) no critical siphon, persistence; (b) a positive steady state exists; (c) at every positive point the Jacobian restricted to the stoichiometric subspace is Hurwitz; (d) every positive class contains exactly one positive steady state, exponentially stable.

Proof. $Y_{\mathrm{e}}(\mathcal{N}^*) = Y_{\mathrm{e}}^{\mathsf{T}}$ has the transposed rank-one blocks on the same clusters, so PC is self-dual; (a) is Theorem 31. (b) With $\theta_c = x^{b_c}$ the rates are $k_t\theta_c^{a_t}$. By Lemma 29 for $\mathcal{N}^*$ with routing $z > 0$ (a conservation law), $\operatorname{im}\Gamma^{\mathsf{T}}$ is spanned by $\Gamma^{\mathsf{T}}(z \odot \mathbf{1}_{S_c})$; so steady states are the zeros of $H_c(\theta) = \sum_{p\in S_c} z_p(\Gamma w(\theta))_p$, and the argument of Theorem 38 applies. (c) $J = PQ$ with $G = QP$ Metzler, irreducible, with the positive null vector $u(a \circ w)$ of Lemma 29. (d) The degree argument of Theorem 39. □

Corollary 43 (block-homogeneous networks; agent Q3). If $\mathcal{N}$ is BH (sources $a\mathbf{1}_B$), connected, with (H1), (H2) and $s = \#\text{blocks} - 1$, then $\mathcal{N}$ has (a)–(d) of Theorem 42, and is globally stable in each class when $s \leq 2$. $\mathcal{N}$ is EC iff the multiples are constant on each block; otherwise it is not EC. The BH networks with these hypotheses are exactly the duals of WF-EC networks.

Proof. If $\mathcal{N}$ is BH, two species of one block have equal rows of Y_{e}, so $\mathcal{N}^*$ is EC with clusters the blocks, connected, of rank s, with (H1), (H2) exchanged and $s = m(\mathcal{N}^*) - 1$: $\mathcal{N}^*$ is WF-EC; conversely the dual of a WF-EC network is BH. Apply Theorems 38 and 39, Corollary 32, and Proposition 40 to the WF-EC network $\mathcal{N}^*$ (or Theorem 42). □

Examples: N_2^* is BH with blocks $\{K_1\}, \{K_2\}$ and not EC; $NP\colon A+2B \to C,\ 2A+4B \to C+D,\ C \to A+2B,\ D \to A+2B$ is PC, neither EC nor BH, connected, with (H1), (H2) and $s = q-1 = 2$. V2 checked Theorem 42 on 4×120 random PC networks; the evidence for genuinely PC networks is thin, and the proof carries the claim.

4.6 The boundary: the cluster rank equation does not give uniqueness

Agent Q3 conjectured that the cluster rank equation alone gives absolute consistency, uniqueness and stability. V2 refuted the dynamical part.

Proposition 44 (the counterexample NC3; V2). $NC3\colon D \to B+C,\ C \to A,\ A+B \to B+C,\ 2A+B \to C+D$ (species A, B, C, D) is connected, satisfies (H1)–(H3) on both sides (laws $B+D$, $A+C+D$; flux rays $(0,1,1,0)$, $(1,2,0,1)$), has $s=2$ and the clusters $\{D\},\{C\},\{A,B\}$, so $q=3=s+1$; its sources sharing a species have the same support, but it is not PC, and neither side has a critical siphon. For $k=(1/6,1/3,6,6)$ the class $B+D=10/3$, $A+C+D=19/2$ contains exactly three positive steady states: $A=1/2$, i.e. $x=(1/2,1/3,6,3)$, a saddle ($\det J|_S=-7/9$), and $A=(7\pm\sqrt{30})/6$, both exponentially stable in the class.

Proof. With $r_1=k_1D$, $r_2=k_2C$, $r_3=k_3AB$, $r_4=k_4A^2B$: $f_B=-f_D=r_1-r_4$ and $f_A+f_C+f_D=0$, so steady states are $r_1=r_4$, $r_2=r_3+2r_4$, i.e. $D=36A^2B$, $C=18AB+36A^2B$ at these rates. The class gives $B=(10/3)/(1+36A^2)>0$ and then $72A^3-204A^2+122A-19=(2A-1)(36A^2-84A+19)=0$. Every positive root gives positive B,C,D, and both roots of the quadratic are positive. The Jacobians on S are computed exactly in $\mathbb{Q}(\sqrt{30})$. (V2 found the three solutions with msolve; this elementary count is ours.) □

A second example of V2, $NC\colon C \to B,\ 2A \to D,\ 2B+D \to 2A+2B,\ 2B+2D \to 2A+2C+D$, with $k=(2/3,5/18,24,64)$, has the saddle $x=(3,1/2,3,1/4)$ ($\det J|_S=-88/9$) in the class $B+C=A+2D=7/2$; the class is compact without boundary steady states (Theorem 31), so by the degree argument of Theorem 39(b) the indices of its steady states add up to $(-1)^s=+1$, and the class contains at least one further positive steady state (V2: exactly two more, by exact root isolation; an exact Gröbner-basis count of reviewer R1 agrees). So *proportional conflict is the boundary of the dynamical statements, while the siphon statement holds on the whole class of the cluster rank equation.* Whether that class gives absolute consistency is open. Without the cluster rank equation every transfer fails, already for small networks (Table 45).

For NM, steady states have $x_3=1/2$, $x_1x_2^2=1/4$, and in the class x_2 is a root of $t^3-2t^2+1/4$ in $(0,2)$; the signs at $-1,0,1,2$ show exactly two.

4.7 Weakly reversible networks have absolutely consistent duals

Theorem 47 (agents E2 and A2). Let $\mathcal{N}$ be conservative with $\operatorname{im}\Gamma \subseteq \operatorname{im} Y_{\mathrm{e}}$ (e.g. every product complex in the span of the sources; in particular $\mathcal{N}$ weakly reversible). Then for every $\kappa \in \mathbb{R}^n_{>0}$, $\mathcal{N}^*$ has a positive steady state. Moreover its positive steady-state set is a smooth manifold of dimension $r-s$, a global graph over a fixed linear space in logarithmic coordinates.

Proof. Existence (E2, Corollary E2.5′): $c^{\mathsf{T}}Y_{\mathrm{e}}=0$ implies $c^{\mathsf{T}}Y_{\mathrm{p}}=0$, so $\ker Y_{\mathrm{e}}^{\mathsf{T}} \subseteq \ker \Gamma^{\mathsf{T}}$; Birch's theorem [12] [11] applied to $\mathcal{N}^*$ gives, for every κ, a positive v with $v-v^0 \in \ker Y_{\mathrm{e}}^{\mathsf{T}}$ and $\log v-\log\kappa \in \operatorname{im} Y_{\mathrm{e}}$, which is a steady-state flux. Manifold (A2): factor $\operatorname{im}\Gamma=\operatorname{im}\Pi$ with $\Pi=Y_{\mathrm{e}}R$; in coordinates $z=q+Rt$, the strictly convex coercive function $F_q(t)=\sum_i \kappa_i e^{(Y_{\mathrm{e}}q)_i}e^{\pi_i\cdot t}$ has a unique minimiser, which depends smoothly on q, and its critical-point equation is the steady-state equation. Details: agent A2's notes

conclusion	network	(m, q, s)
absolute consistency of $\mathcal{N}^*$	$U\colon X_4 \to X_2 + X_3,\ X_3 + X_4 \to X_1,\ X_1 + X_2 \to 2X_4$ has the steady state $(k_1/k_3, 1, k_1/k_2, 1)$ for every k; U^* has none if $\kappa_2 \geq \kappa_1$ ($f_{K_1} - f_{K_2} = (\kappa_2 - \kappa_1)K_3 + 2\kappa_3 K_2$)	$(3, 2, 2)$
persistence of $\mathcal{N}^*$	$3A \rightleftarrows 2A + B$ (Example 50)	$(2, 1, 1)$
same, $s = 3$	$X_4 \to 3X_3 \to X_1 + 2X_2 \to 3X_2 \to X_4$: $\mathcal{N}^*$ loses the species dual to $X_1 + 2X_2 \to 3X_2$	$(4, 3, 3)$
uniqueness; no critical siphon	$NM\colon 2X_3 \to X_1 + X_2,\ 3X_3 \to X_2 + 2X_3,\ X_1 + 2X_2 \to 3X_3$ (PC): two steady states in $x_1 + x_2 + x_3 = 5/2$ for $k = (1, 2, 1)$; critical siphons $\{X_1, X_3\}, \{X_2, X_3\}$	$(3, 2, 2)$
uniqueness (BH)	$NB\colon X_3 \to X_2,\ 2X_3 \to X_1,\ X_1 + X_2 \to 3X_3$: two steady states $x_3 = 1$, $x_1 = (5 \pm \sqrt{17})/4$ in $2x_1 + x_2 + x_3 = 6$ for $k = \mathbf{1}$	$(3, 2, 2)$
absolute consistency of $\mathcal{N}^*$ (EC)	NT of Proposition 41	$(2, 2, 2)$

Table 45 / Conclusions that fail without the cluster rank equation $s = q - 1$. Every network is connected and satisfies (H1)–(H3) on both sides. m: number of sources; q: number of clusters. Agents P, G, E, Q3, with V2's correction of the labelling in the fourth row.

(Theorems E, E+), re-derived by V3 (§6). □

The theorem says nothing about classes: Theorem 70 with $R = 2$ gives weakly reversible $\mathcal{N}$ whose dual loses a species for every rate vector. The convex argument is the classical Birch/moment-map structure (cf. Müller and Regensburger [34]).

5 What the duality does not transfer

5.1 Weak reversibility, deficiency, endotacticity, the siphon criterion

Proposition 48 (agents G, E). The dual of a weakly reversible network need not be weakly reversible: W is weakly reversible with deficiency zero and W^* is not; the smallest conservative example in agent E's exhaustive families is $2X_2 \rightleftarrows X_1$, with dual $K_2 \to K_1,\ 2K_1 \to 2K_2$. If $\mathcal{N}$ is weakly reversible with (H3), the rows of Y_e are pairwise distinct and so are the rows of Y_p, and $\mathcal{N}^*$ is weakly reversible iff there is a permutation π of the species with $(Y_p)_{i\cdot} = (Y_e)_{\pi(i)\cdot}$ for all i; then the deficiency of $\mathcal{N}^*$ is $(n - s) - c(\pi)$, where $c(\pi)$ is the number of cycles of π, and $\mathbf{1}_C$ is a conservation law of $\mathcal{N}$ for every cycle C of π.

Proof. If $\mathcal{N}$ is weakly reversible, every product complex is a source complex, so equal rows of Y_e would give equal rows of Y_p and of Γ, excluded by (H3); the reversed network gives the same for Y_p. So $\mathcal{N}^*$ has n distinct sources and n distinct products. If $\mathcal{N}^*$ is weakly reversible, each product is a source, which defines an injective map π with $(Y_p)_{i\cdot} = (Y_e)_{\pi(i)\cdot}$; conversely the complex graph $i \mapsto \pi(i)$ of a permutation is a disjoint union of cycles. It has n complexes and $c(\pi)$ linkage classes, so the deficiency of $\mathcal{N}^*$ is $n - c(\pi) - s$, and $\sum_{i \in C} \Gamma_{i\cdot} = \sum_{i \in C}((Y_e)_{\pi(i)\cdot} - (Y_e)_{i\cdot}) = 0$ for every cycle C of π. The examples are certified (agent G, Prop. 2.2; V3 §1). □

Blanchini and Giordano already observe that the dual of a reversible network is not

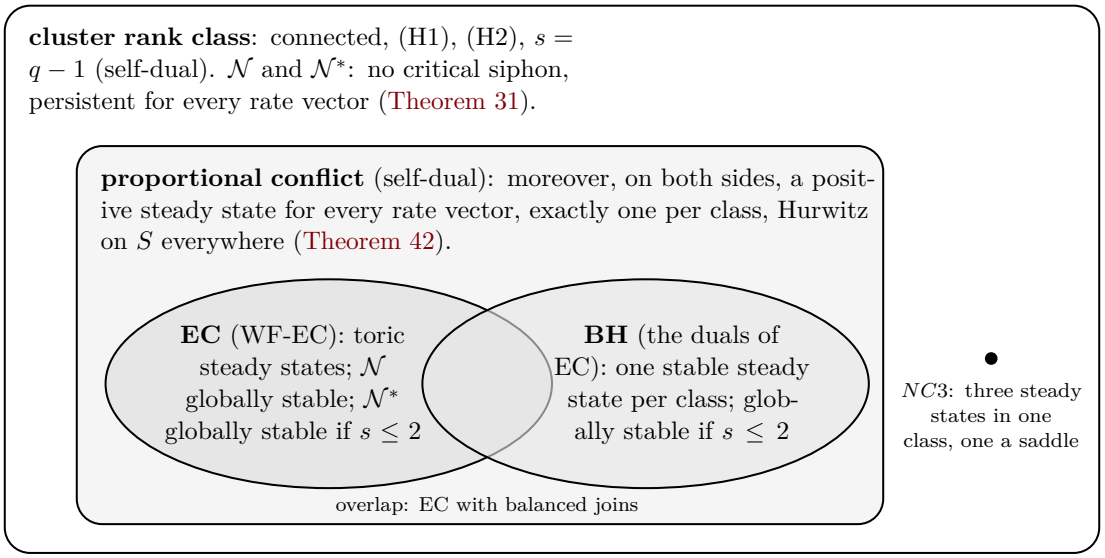


Figure 46 / The classes on which the duality transfers structure (all with connectedness, (H1) and (H2)). Each class inherits the statements of the classes containing it. Global stability of $\mathcal{N}^*$ for $s \geq 3$ with unbalanced joins, and of general PC networks, is open. $NC3$ satisfies the cluster rank equation but not proportional conflict. The networks of Table 45 (U, $3A \rightleftarrows 2A+B$, NM, NB, NT) violate the cluster rank equation and lie outside the outer class.

even weakly reversible and that mass action is not preserved p. 493 [16]. On the 82 377 weakly reversible networks with (H1)–(H3) of agent E's exhaustive families and two seeded samples, the dual is weakly reversible for 98, deficiency zero is preserved for 43 of 46 709, the dual is endotactic for 122 of 84 103 tests and strongly endotactic for 43 of 70 185 (counted per run, about 1 700 networks twice; checked by computation, `gac-duality/computation/notes.md` §5). So the endotactic persistence theorems [13] [14] almost never apply to $\mathcal{N}^*$.

Proposition 49 (the siphon criterion is not self-dual; agent G). $H : 2A \rightleftarrows 2B,\ 2A \rightleftarrows A + B$ is weakly reversible with (H1)–(H3), and its only minimal siphon $\{A, B\}$ supports $a + b$. Its dual $2U_1 + 2U_3 + U_4 \to 2U_2 + U_3 + 2U_4,\ 2U_2 + U_4 \to 2U_1 + U_3$ satisfies (H1)–(H3) and has the critical siphon $\{U_4\}$.

Proof. Checked in exact arithmetic (`certify_duality.py`, claim "S5 H"); agent G, re-checked by V3. □

The converse also fails (agent E: $X_4 \rightleftarrows X_2 + X_3,\ 2X_4 \rightleftarrows X_1 + X_3$); among the 82 377 networks the criterion holds for $\mathcal{N}$ but not $\mathcal{N}^*$ in 5 252 cases and conversely in 924 (checked by computation). In intermediate dimensions almost every weakly reversible network has a critical siphon (agent E), so the criterion rarely applies on either side.

5.2 Persistence

Example 50 (agent G; strengthened by agent E). $M : 3A \rightleftarrows 2A + B$ is reversible with one linkage class and deficiency zero, and $\dot{a} = a^2(k_2 b - k_1 a)$: every class converges to its unique positive equilibrium. Its dual $M^* : 3U + 2V \to 2U + 3V,\ V \to U$ ($U = K_1$, $V = K_2$ in Figure 4) has $\dot{v} = v(\kappa_1 u^3 v - \kappa_2)$, so on a class $u + v = T$, $\dot{v} \leq -(\kappa_2/2)v$ whenever $v \leq \min(T, \kappa_2/(2\kappa_1 T^3))$: M^* is non-persistent for every rate vector and in every class. With unit rates and $T = 2$ the class contains the two positive equilibria

$v = 1$ and a root of $v^3 - 5v^2 + 7v - 1$ in $(1/8, 1/2)$, and $v \to 0$ from $(15/8, 1/8)$ since $h(v) = (2-v)^3 v$ is increasing on $[0, 1/8]$ with $h(1/8) = 3375/4096$, so $\dot{v} \le -(721/4096)v$. Both networks have $\Gamma = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and satisfy (H1)–(H3). ⌟

The smallest instance (V3) is $X : 2X_2 \rightleftarrows X_1 + X_2$, whose dual $K_2 \to K_1$, $2K_1 + K_2 \to K_1 + 2K_2$ satisfies $\dot{K}_2 = K_2(\kappa_2 K_1^2 - \kappa_1)$ and loses K_2 in every class with $\kappa_2 T^2 < \kappa_1$, for every rate vector (Example 62). The mechanism of Example 50 is general.

Proposition 51 (drain species; agent E)**.** Let $\mathcal{X}$ be conservative and i a species such that (a) some reaction has source exactly $d_0 X_i$ and consumes X_i, and (b) every reaction with a net production of X_i ($\Gamma_{ij} > 0$) has a source with X_i-coefficient $> d_0$. Then for every rate vector and every class, every positive solution with $x_i(0)$ small has $x_i(t) \to 0$.

Proof. On a class all coordinates are bounded by some T; the terms of $\dot{x}_i$ with $\Gamma_{ij} > 0$ are $\le C x_i^{d_0+1}$ and the draining reaction contributes $-c_0 x_i^{d_0}$, so $\dot{x}_i \le x_i^{d_0}(C x_i - c_0)$. □

Hypothesis (b) never holds on a weakly reversible network itself (the cycle through the source $d_0 e_i$ must raise the coefficient of X_i to d_0; V3), so every drain species of $\mathcal{N}^*$ is an asymmetry of the duality.

Theorem 52 (drain sets; agent Q1)**.** Let $\mathcal{X}$ be conservative and Z a set of species with weights $w \in \mathbb{R}^Z_{>0}$ and degrees $d \in \mathbb{Z}^Z_{\ge 1}$ such that, with $V = \langle w, x_Z \rangle$: (D1) for each $i \in Z$ some reaction has source exactly $d_i X_i$ and strictly decreases V; (D2) every reaction that strictly increases V has a source y with $\sum_{i \in Z} y_i/d_i > 1$. Then for every rate vector and every class there is $\delta > 0$ such that every solution with $V(x(0)) < \delta$ has $x_Z(t) \to 0$. Z contains no support of a nonnegative conservation law.

Proof. With $u_i = x_i^{d_i}$, $U = \max u_i$ and $\theta = \sum_i \beta_i/d_i > 1$ for a source β of (D2), $x_Z^{\beta} \le U^{\theta} \le U^{\theta-1} \sum_i u_i$, and $U^{\theta-1} \to 0$ as $V \to 0$. So for small V, $\dot{V} \le -(c/2) \sum_i x_i^{d_i}$, and $V \to 0$. □

Example 53 (agents E, E2, Q1)**.** (1) $X_4 \to 3X_3 \to X_1 + 2X_2 \to 3X_2 \to X_4$ is weakly reversible with deficiency zero, $s = 3$, one linkage class and (H1)–(H3), hence globally stable (Anderson [40]; Pantea [41]); its dual satisfies (H1)–(H3) and the species dual to $X_1 + 2X_2 \to 3X_2$ is a drain species. (2) The 5-cycle $X_5 \to 3X_4 \to X_2 + X_4 \to X_1 + 2X_3 \to 3X_3 \to X_5$ ($s = 4$, deficiency zero) has a dual with the drain species dual to $X_1 + 2X_3 \to 3X_3$. (3) For $C : 2X_3 \to X_2 + X_3 \to X_1 + X_3 \to 2X_3$ ($s = 2$, deficiency zero), the dual $K_3 \to K_2$, $K_2 \to K_1$, $2K_1 + K_2 + K_3 \to K_1 + K_2 + 2K_3$ has $V = K_2 + 2K_3$ with $\dot{V} = -\kappa_1 K_3 - \kappa_2 K_2 + 2\kappa_3 K_1^2 K_2 K_3$: $\{K_2, K_3\}$ is a drain set and there is no drain species. (4) Weak reversibility and purity together do not help: $X_4 \rightleftarrows X_2 + X_3$, $3X_3 \rightleftarrows X_1 + 2X_4$ is pure and weakly reversible with (H1)–(H3), and its dual has a drain set of two species (E2; this refutes a conjecture of Q1). ⌟

Counts (checked by computation): among the 82 377 weakly reversible networks with (H1)–(H3) of agent E's families of molecularity at most three (all with $n \le 6$), 141 have a dual with a drain species, all with $s \le 2$; in E's exhaustive molecularity-three families with four and five species, 103 duals with $s = 3$ have one; in E2's seeded molecularity-three samples with $s = 4$, 238 (`gac-duality/computation/notes.md` §5, `gac-duality/round2/computation/notes.md` §3). Drain sets with $|Z| \le 4$ occur for 1 735

duals of E's weakly reversible networks with (H1)–(H3), 1 432 of them without a drain species (Q1, `gac-duality/round2/persistence/notes.md` §3); over all of E's records, 199 duals with $s = 2$ have a drain species, a count recounted by V3.

5.3 Convergence

Persistence of the dual for all rates does not give convergence, even with a unique equilibrium in every class.

Theorem 54 (agent A2). Let $\mathcal{N}$ be the 3-cycle $3X_3 \to X_1 + X_2 \to X_2 + 2X_3 + 3X_4 \to 3X_3$ (weakly reversible, deficiency zero, $s = 2$, (H1)–(H3)). Write a, b, c for the dual species and $\alpha, \beta, \gamma, \delta$ for the dual rates, so that $\mathcal{N}^*\colon b \to a,\ b + c \to a + b,\ 3a + 2c \to 2b + 3c,\ 3c \to 3b$. Then for every rate vector $\mathcal{N}^*$ is persistent (Theorem 68) and has exactly one positive equilibrium in each class $a+b+c = T$. At $(\alpha, \beta, \gamma, \delta) = (17, 400, 70000, 10^6)$ and $T = 201/100$ the equilibrium is $(1, 1, 1/100)$, a hyperbolic source, and the class contains a nonconstant periodic orbit.

Proof. Uniqueness: $\dot{b} - 2\dot{c} = (2\beta c - \alpha)b + 9\delta c^3$, so equilibria have $0 < c < \alpha/(2\beta)$, $b = 9\delta c^3/(\alpha - 2\beta c)$ and $a^3 = 3\delta c(\alpha + \beta c)/(\gamma(\alpha - 2\beta c))$; both are increasing in c (numerators $c^2(3\alpha - 4\beta c)$ and $\alpha^2 + 2\alpha\beta c - 2\beta^2 c^2$), so $a + b + c$ increases from 0 to ∞. At the given rates the fluxes at $(1, 1, 1/100)$ are $(17, 4, 7, 1)$, and the Jacobian in (b, c) is $J = \left(\begin{smallmatrix} -59 & 3658 \\ -25 & 79 \end{smallmatrix}\right)$, trace 20, determinant 86 789, eigenvalues $10 \pm i\sqrt{86\,689}$. With $H = \left(\begin{smallmatrix} 93\,655 & -290\,457 \\ -290\,457 & 13\,471\,234 \end{smallmatrix}\right)$, positive definite, $J^{\mathsf{T}}H + HJ = 3\,471\,560\, I$, so the equilibrium is a repeller and lies in the ω-limit set of no other point. By Theorem 68 every ω-limit set of an interior point is a compact subset of the open triangle, and it contains no equilibrium; Poincaré–Bendixson gives a periodic orbit. □

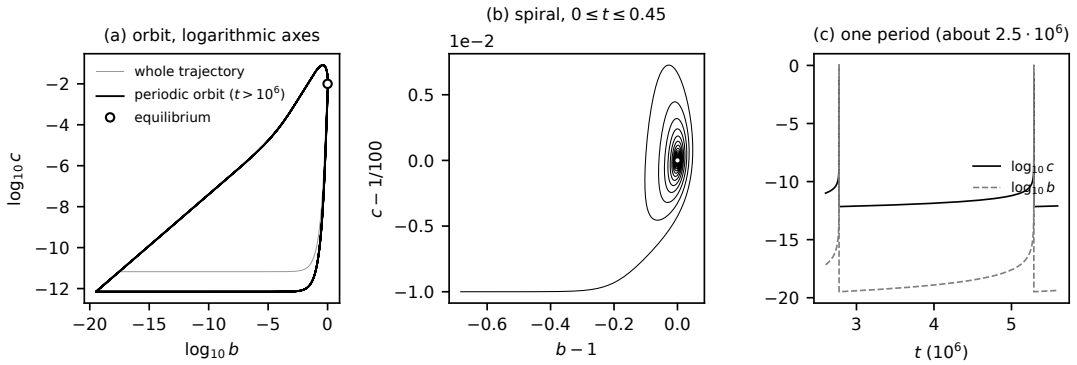


Figure 55 / Numerical illustration of Theorem 54, not part of the proof. (a) The trajectory from $(1, 1 - 10^{-4}, 10^{-2} + 10^{-4})$ in logarithmic coordinates, and its late part, which traces the periodic orbit. (b) The spiral leaving the source. (c) One period of the orbit, about $2.51 \cdot 10^6$ time units, during which c drops to about $7 \cdot 10^{-13}$ and b to about $3 \cdot 10^{-20}$, near the vertex $(T, 0, 0)$. Integrated in the variables $\log a, \log b, \log c$ with a stiff implicit solver (`scripts/figures.py`); in linear coordinates the solver wrongly appears to converge to the vertex (V3).

V3 adds two remarks. The orbit is persistent but passes within about 10^{-12} of the boundary, so naive numerical integration in linear coordinates wrongly suggests extinction: numerical persistence tests on these duals need logarithmic coordinates. And Theorem 54 does not contradict Blanchini and Giordano [16]: their Theorem 1 transfers *structural* local stability (for all admissible derivative matrices) under their Assumption 3 (a rate

depends on x_i iff $\Gamma_{ij} < 0$), which both $\mathcal{N}$ (X_2 is a catalyst in $X_1 + X_2 \to X_2 + 2X_3 + 3X_4$) and $\mathcal{N}^*$ (b is a catalyst in $b + c \to a + b$) violate.

6 Persistence of duals in low rank

6.1 Rank one: an exact test and a dichotomy

Let $\mathcal{M}$ be a conservative network of rank one with reaction vectors $c_j g$, $g \in \mathbb{Z}^p$ primitive, so that g has entries of both signs. A positive class is a segment $x_0 + \xi g$, and $\dot{\xi} = F(\xi) = \sum_j c_j \kappa_j x^{y_j}$. Put $N_- = \{i : g_i < 0\}$ and $N_+ = \{i : g_i > 0\}$. An *upper face* is a nonempty $Z \subseteq N_-$ ($\sigma = +1$), a *lower face* a nonempty $Z \subseteq N_+$ ($\sigma = -1$); every such Z is the zero set of an endpoint of some class. For a face, let $o_Z(j) = \sum_{i \in Z} (y_j)_i$ over the reactions with $c_j \neq 0$, μ its minimum and J the minimisers; $j \in J$ is *inward* if $\sigma c_j < 0$ and *outward* otherwise; β_j is y_j restricted to the species outside Z. The face is IN if all of J is inward, OUT if all is outward, ESC if some outward β_j lies outside the convex hull of the inward ones, and DOM otherwise (Figure 56).

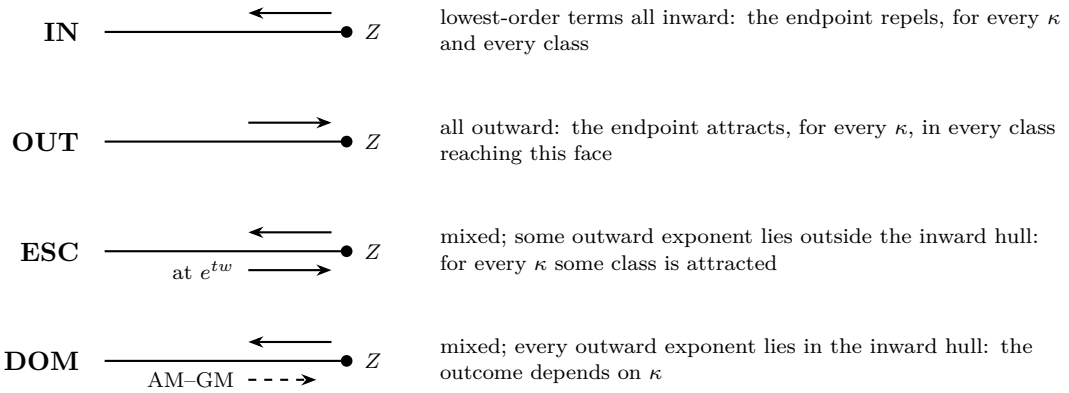


Figure 56 / The four face types of the rank-one test (Theorem 58). Each line is a compatibility class (a segment) ending at a boundary point whose zero set is the face Z; arrows show the lowest-order terms of the flow near that point (left: towards the interior; right: towards the face). For ESC, a separating direction w makes an outward term win at boundary points $\exp(tw)$ for large t; for DOM, every outward exponent lies in the convex hull of the inward ones, so weighted AM–GM bounds it when the outward rate constants are small.

Lemma 57 (endpoint expansion; agent Q1)**.** At an upper endpoint x^+ with zero set Z, along $x^+ - \varepsilon g$,

$$F = \varepsilon^{\mu} L(\kappa, x^+) + O(\varepsilon^{\mu+1}), \qquad L = \sum_{j \in J} c_j \kappa_j \prod_{i \in Z} |g_i|^{(y_j)_i} \prod_{i \notin Z} (x_i^+)^{(y_j)_i};$$

similarly at lower endpoints. $\mathcal{M}$ is persistent for a given κ iff no endpoint of any class attracts nearby points of its class.

Proof. Agent Q1's notes, Lemmas 1.1–1.2 (along a class F is a polynomial in ε, and the dynamics on a segment is one-dimensional); re-derived by V3 (§3). □

Theorem 58 (rank one; agent Q1, with V3's wording of (4))**.** Let $\mathcal{M}$ be conservative of rank one.

(1) $\mathcal{M}$ is persistent for every rate vector iff every face is IN.

(2) If some face is OUT or ESC, $\mathcal{M}$ is non-persistent for every rate vector.

(3) If every face is IN or DOM, some face is DOM, and all DOM faces lie at the same end, the set of rate vectors for which $\mathcal{M}$ is persistent contains a nonempty open set, and so does its complement.

(4) If DOM faces occur at both ends, the set of rate vectors for which $\mathcal{M}$ is persistent is semialgebraic; it can be a hypersurface.

Proof. (1) If all faces are IN, $\sigma L < 0$ always. Otherwise take κ_{j_0} large for an outward j_0 at $x^+ = (0 \text{ on } Z, 1 \text{ off } Z)$. (2) OUT: $\sigma L > 0$ always. ESC: a vector w separating an outward β_0 from the inward hull; at $x^+ = \exp(tw)$ off Z the outward term dominates as $t \to \infty$. (3) Weighted AM–GM bounds each outward monomial by the inward ones; small outward rate constants make $\sigma L < 0$ everywhere, by finitely many strict linear inequalities in κ. (4) The condition is a first-order formula in κ (Tarski–Seidenberg); example B below. □

For a rank-one network $\mathcal{N}$ with reaction vectors $b_j a$, $\Gamma(\mathcal{N}^*) = ba^{\mathsf{T}}$: a face of $\mathcal{N}^*$ is a nonempty set Z of reactions of $\mathcal{N}$ all oriented the same way along a, the order of ρ_i is the *load* $\ell_Z(i) = \sum_{j \in Z} (y_j)_i$, and ρ_i is inward iff X_i is a net product of Z.

Corollary 59 (the test on $\mathcal{N}$; agent Q1)**.** For $\mathcal{N}$ of rank one with (H2), $\mathcal{N}^*$ is persistent for every rate vector iff for every nonempty set Z of reactions with a common orientation, every species with $a_i \neq 0$ minimising ℓ_Z is a net product of Z.

Proof. Theorem 58(1) for $\mathcal{M} = \mathcal{N}^*$, which is conservative by (H2), with the dictionary above. □

Theorem 60 (all-or-none in rank one; agent Q1)**.** Let $\mathcal{N}$ have rank one and suppose that in every linkage class the two complexes extreme along a are sources (e.g. $\mathcal{N}$ weakly reversible). Then $\mathcal{N}^*$ has no DOM face, so it is persistent for every rate vector or for none.

Proof. A linkage class lies on a line $y_L + ta$. Suppose a forward face Z is DOM, with outward species o and inward species q of minimal load, and $\beta_o = \sum_q \lambda_q \beta_q$. Then $\phi_o - \phi_v$ ($v = \sum \lambda_q X_q$) is affine in t with negative slope and vanishes at the sources of reactions outside Z, so at most one position per linkage class is such a source; the extreme forward position is one, every reaction of Z leaves a position where $\phi_o - \phi_v > 0$, and summing over Z contradicts the equality of the minimal loads. □

Corollary 61 (agent Q1, with V3's correction)**.** If $s = 1$, $\mathcal{N}$ satisfies (H1) and (H2), and no reaction of $\mathcal{N}$ has one of its net products in its source (e.g. $\mathcal{N}$ pure), $\mathcal{N}^*$ is persistent for every rate vector. (H1) is needed: $0 \rightleftarrows A$ is pure, weakly reversible and consistent, and its dual $K_2 \to K_1$ loses K_2. If $n = 2$, $s = 1$ and (H1)–(H3) hold, $\mathcal{N}^*$ is persistent for every rate vector iff every source contains strictly more of the species its reaction consumes than of the one it produces.

Proof. By (H1) the vector a has entries of both signs. Let Z be a face of $\mathcal{N}^*$ (Corollary 59). A species with $a_i \neq 0$ is a net reactant of every reaction of Z or a net product of every reaction of Z. A net reactant occurs in every source of Z, so its load is at least $|Z| \geq 1$; a net product occurs in no source of Z by hypothesis, so its load is 0, and net products exist by (H1). So every species of minimal load with $a_i \neq 0$ is a net product, every face is IN, and Corollary 59 applies. If $n = 2$, every reaction consumes one species and produces the other, the reactions of one orientation all consume the same species, loads add over Z, so the singleton faces decide, and $\{j\}$ is IN iff y_j contains more of the consumed than of the produced species. □

Example 62 (rank one; agents Q1, E2, V3). $X : 2X_2 \rightleftarrows X_1 + X_2$ (reversible, deficiency zero, (H1)–(H3)): the face $\{K_2\}$ of X^* is ESC, so X^* is NONE although it has no drain species and no drain set; it is the smallest weakly reversible network with (H1)–(H3) and a NONE dual (V3: one species is impossible, $X_1 \rightleftarrows X_2$ is ALL, and V3's enumeration of the family $(2, 2, 4)$ finds X first). For $3A \rightleftarrows 2A+B$ the face $\{K_2\}$ is OUT. $P : 2X_1 + X_3 \rightleftarrows 3X_1 + 2X_2,\ 2X_2 + X_3 \rightleftarrows X_1 + 4X_2$ (weakly reversible, $s = 1$): every singleton face of P^* is IN, but the face formed by the two forward reactions is ESC, so P^* is NONE. $S : X_1 + X_2 \to 2X_2 + 2X_3,\ X_1 + X_2 + 2X_3 \to 2X_1$ (not weakly reversible): $\dot{K}_1 = K_2((\kappa_2 - \kappa_1)K_1 + 2\kappa_3 K_2)$ for $S^* : K_1 + K_2 \to 2K_2,\ K_1 + K_2 \to 2K_1,\ 2K_2 \to 2K_1$, persistent exactly for $\kappa_1 > \kappa_2$ (DOM at one end). $B : X_1 + X_2 \to 2X_2,\ X_1 + X_2 \to 2X_1$: $\dot{K}_1 = (\kappa_2 - \kappa_1)K_1K_2$, persistent only on $\kappa_1 = \kappa_2$ (DOM at both ends). In rank two the corollary for pure networks fails, already without weak reversibility: $Q : X_1 \to X_2,\ X_1 \to X_3,\ X_2 + X_3 \to 2X_1$ has $(K_2 - K_1)^{\cdot} = (\kappa_3 - \kappa_2)K_3$, so Q^* is non-persistent for every rate vector with $\kappa_2 \neq \kappa_3$ (the hypersurface $\kappa_2 = \kappa_3$ was not analysed); Example 53(4) shows that it also fails for weakly reversible pure networks. ⌟

Counts and credits. On eight exhaustive families of weakly reversible rank-one networks with (H1)–(H3) (up to 3 544 networks per family), $\mathcal{N}^*$ is persistent for every rate vector for 11–33 % of the networks of each family with more than one member (381 of 3 544 to 4 of 12; the family $(4, 3, 4)$ has a single network, which is ALL) and for none for the rest, never SOME (Q1; recounted by V3 with its own generator, identical numbers). E2's separate decision procedure agrees on all 4 754 rank-one networks with (H1), (H2) of E's records. The face test restricts the parallel-sweep test of Craciun, Nazarov and Pantea [13] and Pantea [41] to the directions $\mathbf{1}_Z$. Zhang, Gao and Dochain Thm 3.10 [42] prove persistence of one-dimensional W_I-endotactic networks for all rates, which implies the "if" half of Theorem 58(1) under a stronger hypothesis: V3's network $A + B \to 2A,\ 2A \to A + B,\ 3A + B \to 4A$ has every face IN ($\dot{a} = a(k_1 b - k_2 a + k_3 a^2 b)$) but is not W_I-endotactic. The converse halves, the DOM case and the dichotomy were not found.

6.2 Weak reversibility excludes dominated facets in every rank

Theorem 63 (agent D2, in V3's general form). Let Z be a set of reactions of $\mathcal{N}$ with positively parallel reaction vectors $b_j a$ (a facet set of $\mathcal{N}^*$, in any rank), and suppose every product of a reaction of Z is a source complex (e.g. $\mathcal{N}$ weakly reversible). Then the facet Z of $\mathcal{N}^*$ is not DOM: there are no outward species o and inward species q of minimal load with $(y_l)_o = \sum_q \lambda_q (y_l)_q$ for every $l \notin Z$, with $\lambda_q \geq 0$ and $\sum_q \lambda_q = 1$.

Proof. $\phi(y) = y_o - \sum_q \lambda_q y_q$ vanishes on the sources outside Z and drops along each reaction of Z. A source of Z with $\phi \le 0$ would start an infinite chain of distinct complexes with strictly decreasing ϕ. So $\phi > 0$ on the sources of Z, contradicting $\sum_{j\in Z} \phi(y_j) = \mu - \mu = 0$. □

6.3 Rank two

Let $s = 2$ and let $\mathcal{N}$ satisfy (H1)–(H3), so that the positive classes of $\mathcal{N}^*$ are compact convex polygons. Factor $\Gamma = \Pi\hat{G}$; the class of K^0 is the polygon $\{X \in \mathbb{R}^2 : K^0_j + \langle g_j, X\rangle \ge 0\}$, whose edges have inward normals among the reaction vectors g_j of $\mathcal{N}$. Every point with zero set exactly a facet set lies in the relative interior of an edge of some class, and every point with zero set a vertex set is a vertex of some class (agent D2, Lemma 2.1). On an open edge whose zero set Z is a siphon of $\mathcal{N}^*$ (a *siphon edge*; its facet set is a *siphon facet*) only the reactions of $\mathcal{N}^*$ whose source avoids Z act, and they move along the edge; their sum is the *edge flow*, *one-way* if all its terms have one sign, *two-way* if both signs occur, *zero* if there is no such reaction. The finer notions used in Proposition 66 (two-coloured edge flows, the transverse order μ, tropically Hurwitz vertices, centre manifolds, attracting linear functions and sectors) are defined in agent D2's notes (`gac-duality/round3/dichotomy/notes.md` §2–3).

Lemma 64 (agent D2)**.** If p is a boundary point of the ω-limit set Ω of an interior point, then p is an equilibrium or lies in an open edge whose zero set is a siphon of $\mathcal{N}^*$, the field is tangent to the edge there, and the whole orbit of p stays in the open edge. Every vertex in Ω is an equilibrium.

Proof. Agent D2's notes, Lemma A (the siphon edge is invariant in both time directions for the polynomial field); re-derived by V3 (§5). □

Theorem 65 (persistence certificate; agent D2, corrected by V3)**.** Suppose that for every rate vector and every class: (A1) every siphon facet has a one-way edge flow, or is IN with a two-way edge flow that cannot vanish identically; (A2) every vertex equilibrium e has a neighbourhood U_e and a C^1 function $V_e \ge 0$ with $V_e(e) = 0$, $V_e > 0$ on the interior and $\dot{V}_e > 0$ on $U_e \cap \operatorname{int} P$; (A3) some edge is not a siphon edge. Then $\mathcal{N}^*$ is persistent for every rate vector.

Proof (sketch)**.** A boundary point of Ω leads (Lemma 64) to a boundary equilibrium $e \in \Omega$, which repels the interior by (A2) or by the normal coordinate of an IN edge, and is isolated: along an interior total orbit in a compact invariant set near e, V_e is strictly increasing, so its α- and ω-limit sets lie in ∂P; near e the only invariant subset of ∂P is $\{e\}$ (boundary orbits are monotone along the edges), so both limit sets are $\{e\}$, which contradicts the strict increase of V_e between them since $V_e(e) = 0$ (V3's addition). The Butler–McGehee lemma Thm 4.1 [43] gives a point of $\Omega\backslash\{e\}$ converging to e, necessarily on a siphon edge flowing into e; repeating upstream around the boundary circle makes every edge a siphon edge, contradicting (A3). This is a planar version of the acyclicity theorem of Butler and Waltman [43] and Hale and Waltman [44], with Lemma 64 replacing their invariant-boundary hypothesis. Full proof in agent D2's notes, re-derived by V3. □

Proposition 66 (extinction certificates; agent D2, with V3's repair)**.** Each of the following gives, for every rate vector, a class with an interior trajectory converging to the boundary: (E) an OUT or ESC siphon facet whose edge flow is two-coloured with minimal transverse order $\mu = 1$, or zero with $\mu = 1$ (an edge equilibrium with an attracting normal eigenvalue, whose stable manifold enters the interior; V3 repaired the written proof for $\mu = 1$, which covers every use); (H) a tropically Hurwitz vertex; (CM) an attracting centre manifold at a vertex; (L) an attracting linear function; (S) an attracting sector; (B) a product of two rank-one blocks, one of them NONE.

Proof (sketch)**.** Proofs in agent D2's notes (Propositions E, H, CM, L, S, B; Table 76). (E) for $\mu = 1$, as repaired by V3: on the facet there is a point p^* with zero set Z, zero edge flow and normal coefficient $L(p^*) < 0$; it is an equilibrium of its class, and the stable manifold of the eigenvalue $L(p^*)$ is transverse to the invariant edge and enters the interior. (H): Cauchy–Binet for a sum of rank-one 2×2 matrices; a Hurwitz vertex attracts an open set of its class. (B): the two blocks decouple. (CM), (L), (S) are standard centre-manifold, Lyapunov and sector arguments, written as sketches in D2's notes and checked by V3 at sketch level and on the worked examples below. □

Examples (certified identities): for $2X_3 \to X_2 \to 2X_1 + X_3 \to 2X_3$ the dual $2K_3 \to 2K_2,\ K_2 \to K_1,\ 2K_1 + K_3 \to K_2 + 2K_3$ has $\dot{K}_3 = K_3(\kappa_3 K_1^2 - 2\kappa_1 K_3)$, the only siphon edge is $\{K_3\}$ (IN) and the only siphon vertex $\{K_2, K_3\}$: ALL. For $2X_3 \to 2X_2 + X_3 \to X_1 + X_2 + X_3 \to 2X_3$ the facet $\{K_3\}$ is ESC with zero edge flow and $\mu = 1$: NONE by (E). For $2X_3 \to X_2 + X_3 \to X_1 + X_2 \to 2X_3$, at $(0, T, 0)$ the reduced equation on the centre manifold has the coefficient $\kappa_3 T(\kappa_2 T - \kappa_1)/(\kappa_1 + \kappa_2 T)$: NONE by (CM). For $2X_3 \to 2X_2 + X_3 \to 2X_1 + X_3 \to 2X_3$, $V = K_2 + 2K_3$ has $\dot{V} = -2\kappa_1 K_3^2 - 2\kappa_2 K_2^2 + 2\kappa_3 K_1^2 K_2 K_3$, negative definite near $(T, 0, 0)$ when $\kappa_3^2 T^4 < 4\kappa_1\kappa_2$: NONE by (L). For $2X_3 \to X_2 \to X_1 + X_3 \to 2X_3$ the vertex $(T, 0, 0)$ of the dual has the linearisation $\left(\begin{smallmatrix} -\kappa_2 & \kappa_1 + \kappa_3 T^2 \\ 0 & \kappa_3 T^2 - \kappa_1 \end{smallmatrix}\right)$, Hurwitz when $\kappa_3 T^2 < \kappa_1$ (E2's vertex scaling).

Counts (checked by computation). Among the 18 861 weakly reversible networks with (H1)–(H3) and $s = 2$ in agent E's records, E2 certified 2 988 ALL (no critical siphon in $\mathcal{N}^*$) and 1 048 NONE (drain species, drain sets, vertex scaling); D2's certificates decide 14 654 of the remaining 14 825: in all, ALL 4 873, NONE 13 817, undecided 171, and no network meets an ALL and a NONE certificate (`gac-duality/round3/dichotomy/notes.md` §4; re-run by V3, whose own sampled test of the ALL verdicts found no Hurwitz boundary vertex). The undecided ones are mostly ties in κ at a vertex, as in the next example.

Example 67 (a vertex tie; agent D2)**.** For $3X_3 \to 2X_1 + X_2 \to X_2 + 2X_3 \to 3X_3$, with the reactions numbered $3X_3 \to 2X_1 + X_2$, $X_2 + 2X_3 \to 3X_3$, $2X_1 + X_2 \to X_2 + 2X_3$, the dual $2K_3 \to 2K_1,\ K_2 + K_3 \to K_1 + K_3,\ 3K_1 + 2K_2 \to 3K_2 + 2K_3$ satisfies, with $a = K_2$, $b = K_3$, $K = \kappa_3 K_1^3$, $z = a/b$ and $u = b/a$, $\dot{z} = a[(2\kappa_1 - \kappa_2) + Kz(1 - 2z)]$, $\dot{u} = a[(\kappa_2 - 2\kappa_1)u^2 - Ku + 2K]$, $\dot{a} = a^2(K - \kappa_2 u)$ and $\dot{b} = b^2(-2\kappa_1 + 2Kz^2)$. The sign of $2\kappa_1 - \kappa_2$ at the vertex $(T, 0, 0)$ cannot be changed by scaling the class, and the dual is NONE by three different mechanisms for $\kappa_2 > 2\kappa_1$, $<$ and $=$. ⌟

Conjecture F is false. Q1 and D2 conjectured that a realisable OUT facet forces NONE (D2 proved it for zero edge flows with $\mu = 1$, and for two-coloured edge flows with $\mu = 1$ with V3's repair). Agent A2 refuted it.

Theorem 68 (agent A2). The dual of $3X_3 \to X_1 + X_2 \to X_2 + 2X_3 + 3X_4 \to 3X_3$ (Theorem 54) is persistent for every rate vector, although every positive class has the OUT facet $c = 0$.

Proof. $\mathcal{N}$ is weakly reversible with deficiency zero, rank two and (H1)–(H3) (law $(11, 1, 4, 1)$, value 12 on every complex; flux $(1, 1, 1)$). The dual field is $\dot{a} = \alpha b + \beta bc - 3\gamma a^3c^2$, $\dot{b} = -\alpha b + 2\gamma a^3c^2 + 3\delta c^3$, $\dot{c} = -\beta bc + \gamma a^3c^2 - 3\delta c^3$, on the triangles $a+b+c = T$. On the open edges $a = 0$ and $b = 0$ the field points strictly inward, at $(0, T, 0)$ and $(0, 0, T)$ it is nonzero, and on $c = 0$ the orbit $b(t) = b(0)e^{-\alpha t}$ leaves the triangle in finite backward time; so $\Omega \cap \partial P \subseteq \{(T, 0, 0)\}$. The edge $c = 0$ is OUT: the only term of order one in c is $-\beta bc$. With $V = c\, e^{-\beta b/\alpha}$,

$$\alpha\, \dot{c} - \beta c\, \dot{b} = c^2\big[\alpha\gamma a^3 - (3\alpha\delta + 2\beta\gamma a^3)c - 3\beta\delta c^2\big], \tag{69}$$

so $(\log V)^{\cdot} = c\,[\gamma a^3 - (3\delta + 2\beta\gamma a^3/\alpha)c - 3\beta\delta c^2/\alpha] \geq c\,(\gamma T^3/8 - E\varepsilon) \geq c\gamma T^3/16 > 0$ at the interior points of $\{b + c < \varepsilon\}$, with $E = 3\delta + 2\beta\gamma T^3/\alpha + 3\beta\delta/\alpha$, $\varepsilon = \min(T/2, 1, \gamma T^3/(16E))$. So V increases along interior trajectories while they stay in this region. If $x(t_j) \to (T, 0, 0)$ along an interior trajectory, the trajectory cannot stay in the region from some time on (V would increase while $V(x(t_j)) \to 0$); the last entry points $x(s_j)$ before t_j satisfy $b + c = \varepsilon$ and $V(x(s_j)) \leq V(x(t_j)) \to 0$, so they accumulate at $(T - \varepsilon, \varepsilon, 0) \in \Omega$, a point of the open edge $c = 0$, which was excluded. □

Theorem 70 (agent A2). For integers $u \geq 3$, $R \geq 1$, $N_{u,R}\colon uX_3 \to X_1 + X_2 \to X_2 + 2X_3 + RX_4 \to uX_3$ satisfies (H1)–(H3) (law $(uR + 2, u - 2, R + 1, u - 2)$), is weakly reversible with deficiency zero and rank two, and its dual $b \to a$, $b + c \to a + b$, $ua + 2c \to 2b + uc$, $Rc \to Rb$ has the OUT facet $c = 0$. The dual is ALL for $R \geq 3$ and NONE for $R = 1, 2$.

Proof (sketch). $R \geq 3$: as for Theorem 68, with $(\log V)^{\cdot} = c[(u-2)\gamma a^u - R\delta c^{R-2} - (2\beta\gamma/\alpha)a^u c - (R\beta\delta/\alpha)c^{R-1}]$. $R = 2$: $\dot{c} \leq -\delta c^2$ on the class $T = \min(1, \delta/((u-2)\gamma))$. $R = 1$: $\dot{c} \leq -(\delta/2)c$ on $T = \min(1, \delta/(2(u-2)\gamma))$ (using $T^u \leq T$ for $T \leq 1$). Certified for $u = 3, \ldots, 7$, $R = 1, \ldots, 7$. Full proof in agent A2's notes (Theorem AF), re-derived by V3 (§6). □

Proposition 71 (agent A2). If $\mathcal{N}$ has one linkage class and rank two, every siphon edge of $\mathcal{N}^*$ has a one-way or zero edge flow, for every rate vector.

Proof. Agent A2's notes, Proposition L (opposite active vectors on a siphon edge would give a nonnegative conservation law of $\mathcal{N}$ vanishing on a complex, hence on the whole linkage class); re-derived by V3 (§6). □

Status of the dichotomy. For weakly reversible $\mathcal{N}$ with $s = 2$, every decided network is ALL or NONE, Theorem 63 removes the rank-one mechanism for rate dependence, and no SOME network is known; but vertex ties (Example 67) can make the mechanism depend on κ, and the all-or-none dichotomy at $s = 2$ is open. Outside weak reversibility SOME occurs already at $s = 1$ (Example 62).

7 The Global Attractor Conjecture

Status in October 2026. The GAC asserts that every complex-balanced equilibrium of a mass-action system attracts every positive point of its class. It is claimed in full by a major revision of Craciun's proof [9] (arXiv v3, 23 September 2026) and by OpenAI's preprint on boundedness and persistence of weakly reversible mass-action systems [10] (25 September 2026), from which the GAC follows by the classical entropy argument; the latter comes with a Lean statement of its main theorem. Wiuf [45] proves a special case under siphon conditions. We have not checked these preprints; none mentions the dual network. Classical cases include one linkage class [40], dimension at most three [41], strongly endotactic networks [14] and facets [46]; the source's "small dimensions" refers to [2].

Proposition 72 (the duality keeps the dimension; agents G, E)**.** $\mathcal{N}^*$ has the same rank s as $\mathcal{N}$, and the codimension $n-s$ of $\mathcal{N}$ is the dimension of the cycle space $\ker\Gamma(\mathcal{N}^*)$, not a dimension of $\mathcal{N}^*$. With β^* the cycle nullity of the complex graph of $\mathcal{N}^*$ and δ^* its deficiency, $n-s=\beta^*+\delta^*$; if $n-s=1$ and $\mathcal{N}^*$ is weakly reversible, $\mathcal{N}^*$ is a single directed cycle of deficiency zero. The monomolecular n-cycle has $n-s=1$, $s=n-1$, and a monomolecular n-cycle as dual. No rank-preserving operation (transposition, reversal, relabelling, cancelling common factors, diagonal scaling) turns small codimension into small dimension.

Proof. $\Gamma(\mathcal{N}^*)=\Gamma^{\mathsf{T}}$; $\Gamma(\mathcal{N}^*)=Y^*I^*$ gives $\delta^*=\operatorname{rank}I^*-s$ and $\beta^*=n-\operatorname{rank}I^*$. □

So "small codimension through duality" cannot be the transposition of a small-dimension argument. Three readings of "small codimension" were examined by agent L: the codimension $n-s$ (the cycle space of $\mathcal{N}^*$, as above); the codimension δ of the toric locus in the parameter space of $\mathcal{N}$ Thm 9 [2], which Iosif's framing makes the state space of $\mathcal{N}^*$; and the codimension of boundary faces, where facets repel for weakly reversible networks (Anderson and Shiu [46]). None of them was developed in the literature in connection with $\mathcal{N}^*$, and Section 5 shows that no unconditional transfer of persistence or convergence exists.

Proposition 73 (a codimension-one certificate; agent G)**.** If $\mathcal{N}$ is weakly reversible with (H3), no zero reaction, $n-s=1$, $\mathcal{N}^*$ weakly reversible and n prime, then $\mathcal{N}$ is strongly endotactic, hence permanent for every rate vector [14]. Primality cannot be dropped ($A\to B\to C\to D\to A$, $A+C\rightleftarrows B+D$).

Proof (sketch)**.** By Proposition 48 a single n-cycle of species maps every source to its product; each linkage class is an orbit of the cyclic shift, and since the cyclotomic polynomial Φ_n is irreducible for prime n each class spans S; a direction not orthogonal to S is nonconstant on every class, which gives strong endotacticity. Full proof in agent G's notes (Proposition 4.1), re-derived by V3 (§1). □

The five-species network $X_i+X_{i+2}\to X_{i+1}+X_{i+3}$ together with its double (indices mod 5) satisfies the hypotheses with $s=4$ and deficiency 4, and has the critical siphon $\{X_1,X_2,X_3\}$, realised at $(0,0,0,2,3)$ with face codimension 3; it lies outside the single-linkage, rank-three, siphon and facet/vertex criteria. This is a recognition criterion for a known class, not a new case of the GAC.

Proposition 74 (boundary conditions in dual terms; agent G)**.** (1) Every siphon of $\mathcal{N}$ contains a semiflow support iff every T-trap of $\mathcal{N}^*$ contains an internal cycle; under (H1) either implies persistence of $\mathcal{N}$ for every rate vector [8]. (2) If a face $\{x_\Sigma = 0\}$ of a class contains a point positive off Σ, its codimension is $\operatorname{rank}\Gamma[\Sigma,\cdot]$, the rank of the subnetwork of $\mathcal{N}^*$ with the reactions Σ: facets correspond to rank-one T-traps and vertices to rank-s T-traps, and the facet/vertex criterion of Anderson and Shiu Thm 4.6 [46] reads "every realisable T-trap of $\mathcal{N}^*$ spans rank 1 or s". (3) For weakly reversible $\mathcal{N}$, the hypotheses of Wiuf's Theorem 7.2 [45] read $\operatorname{rank} M[J_0,\cdot] + \operatorname{rank} M[\cdot,\Sigma] = s$ and $\ker M[J_L,\Sigma] \cap \mathbb{R}^\Sigma_{>0} = \emptyset$ with $M = \Gamma(\mathcal{N}^*)$ (V3 checked the reading of condition ($\diamond$) on the PDF).

Proof. Agent G's notes, Propositions 6.1–6.3 ((2): the tangent space of the face is the kernel of the projection of S onto $\mathbb{R}^\Sigma$); re-derived by V3 (§1). □

These are re-expressions of theorems about $\mathcal{N}$, not transfers of dynamics from $\mathcal{N}^*$. The Lyapunov route of Blanchini and Giordano transfers structural local stability, which complex balance does not imply:

Example 75 (agent G)**.** The 3-cycle $3A+2C \to 2A+3B \to 2B+3C \to 3A+2C$ has rank two, deficiency zero, a weakly reversible dual of deficiency zero, and $(Y_e)_{ij} > 0$ iff $\Gamma_{ij} < 0$. The network $10A + 2C \to 9A + 3B,\ 2A + B \to 3C,\ 10B + C \to 3A + 8B$ has the same Γ; at $(1,1,1)$ its derivative matrix is $G = \left(\begin{smallmatrix} 10 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 10 & 1 \end{smallmatrix}\right)$, and ΓG restricted to S is $\left(\begin{smallmatrix} -15 & 27 \\ 24 & -25 \end{smallmatrix}\right)$ with determinant -273. So the structural Jacobian class of a complex-balanced network need not be Hurwitz, and no common polyhedral Lyapunov norm exists. ⌟

Answer to Q3. The duality transfers no known case of the GAC: it keeps the dimension, it leaves the weakly reversible class, and outside the classes of Section 4 it transfers neither persistence nor convergence. With the GAC claimed in full, the meaningful content is structural: the self-dual classes of Figure 46, on which persistence, existence, uniqueness and local stability transfer, and the boundary dictionary above.

8 Open problems

(1) *Dichotomy at $s = 2$.* Is the dual of a weakly reversible network with (H1)–(H3) and $s = 2$ persistent for all rate vectors or for none? (171 networks undecided; vertex ties as in Example 67.)

(2) *Global stability of the dual for $s \geq 3$.* Is $\mathcal{N}^*$ globally stable for WF-EC $\mathcal{N}$ with unbalanced joins and $s \geq 3$? Simulations (Q3, V2, 140 networks) found convergence; the Hurwitz property alone does not suffice (Markus–Yamabe).

(3) *Global stability for proportional conflict.* Do well-formed PC networks have the global attractor property (Theorem 37 uses equal conflict)?

(4) *Absolute consistency under the cluster rank equation.* Does every connected network with (H1), (H2) and $s = q - 1$ have a positive steady state for every rate vector? ($NC3$ and NC do.)

(5) *Absolute consistency of the dual for unbalanced EC networks with $s = m$.* Both cases occur (Proposition 41); characterise them.

(6) *Liveness of weighted impure equal-conflict nets.* The siphon statement is settled by Theorem 31 and Corollary 32; a liveness theory for weighted impure EC nets (Teruel–Silva assume purity) is not.

(7) *An exact algorithm for* $\widehat{\mathsf{P}}$. $\widehat{\mathsf{P}}$ is a meet over a semialgebraic family; a sound lower bound and a sampled upper bound agree on 63 717 of 63 722 tested network sides (Q2), but no finite exact procedure is proved when $s \geq 2$.

(8) *Finest invariant repair.* Is $\widehat{\mathsf{P}}$ the finest duplication- and splitting-invariant coarsening of Definition 19?

(9) *Conjecture F in higher rank and one-way flows.* Which OUT facets force extinction (Theorem 70 shows the answer depends on the network, not only on the facet type)?

9 Computer verification, verification record and credits

9.1 Certificates

All certificates use exact rational arithmetic, print PASS or FAIL per claim, exit with a nonzero status on failure, and state their environment and runtime at the top of their logs.

- `scripts/certify_duality.py` (standard library and `scripts/crn.py`; about 1 s): 130 claims, all PASS (`certificates/certify_duality.log`). It certifies every example of this paper and every finite claim about an explicit network: the dictionary on nine networks, every entry used from Table 10, the hierarchy witnesses and sign-definite one-block certificates for $\widehat{\mathsf{P}}$, the invariance examples, Theorem 21's degenerate case, the K8 pairs with all relabellings, the hypotheses and steady states of $N_0, N_1, N_2, NT, NT_c, U, NP$, the exact count and stability of the three steady states of $NC3$ in $\mathbb{Q}(\sqrt{30})$, the saddle of NC, the root counts of NM and NB (in $\mathbb{Q}(\sqrt{17})$), the examples of Proposition 49 and of its converse, the extinction identities and drain certificates, the rank-one face types, the rank-two identities, Theorem 68, Theorem 70 for $u \leq 7$, $R \leq 7$, the exact source certificate of Theorem 54, and the examples of Section 7. Polynomial identities are checked with an exact sparse-polynomial class. Run: `py -3 problems/iosif-siphon-precluster-duality/scripts/certify_duality.py` from the repository root.

- `scripts/figures.py` (numpy, scipy, matplotlib; WSL; about 16 s): the numerical illustration Figure 55 (`certificates/figures.log`). It is not a certificate.

- `scripts/certify_examples.py` and the enumeration scripts of the archived report (142 claims; 15 513 networks).

- The large enumerations of the working notes (agents E, E2, Q1, Q2, Q3, D2, P) and the verifiers' recounts are kept in `gac-duality/` with their scripts and logs; the paper cites their counts as "checked by computation" with the path of the note.

9.2 Verification record

The archived report (exact criteria and counterexamples) was verified by GPT-6-Astra (effort xhigh) on 25 October 2026 (`verification.md`). The results of the working notes were re-derived by three verifiers, Claude Opus 5.5 (effort xhigh), each with a fresh context and its own exact library (`gac-duality/verification/V1..V3/`):

- V1: the Petri dictionary, transversality, Theorem 21, the WF-FC structure, K8, and the corrected preclusters (Section 3). No claim refuted; corrections applied here: the definition of twins in Theorem 21; the hypotheses of the Petri-cluster correspondence; "always" replaced by "for every consistent (conservative) network" in Theorem 19; the reading of Conradi–Flockerzi in Lemma 14; "canonical" dropped; the species-side restriction of E's summary sentence; the caveats on K8 (Q3's pairs M_3, M_4, used here, satisfy (H1)–(H3) on both sides). V1's citation correction (Desel–Esparza Thm 6.17) concerns P's liveness statement D(h), not included here.
- V2: the equal-conflict theory, Theorem 31 to Corollary 43, and E2's Propositions E2.3–E2.5. No proof marked proved was found wrong; V2 refuted Q3's Conjecture CR (Proposition 44), corrected the cluster-kinetics remark, the labelling of a row of Table 45 and the critical siphons of NM, and asked for an algebraic definition of WF-EC.
- V3: persistence and obstructions (agents G, E, Q1, E2, D2, A2), Sections 5 to 7. Corrections applied: (H1) in Corollary 61; the wording of Theorem 58(4); $V_e(e) = 0$ and the isolation step in Theorem 65; the proof of Proposition E restricted to $\mu = 1$; the general form of Theorem 63; Conjecture F refuted by Theorem 68; the existence part of Theorem 47 credited to E2.

Items marked "sketch" (Theorems 24, 65 and 70 and Propositions 66 and 73) have complete proofs in the working notes, re-derived by the verifiers, except the tropical certificates (CM), (L), (S) of Proposition 66, whose proofs in D2's notes are themselves sketches; V3 checked them at sketch level and on their worked examples only.

The rewritten manuscript was then reviewed by two fresh-context reviewers, Claude Opus 5.5 (effort xhigh): R1 (mathematics) and R2 (attribution, consistency and presentation); their corrections are applied here (`gac-duality/review/R1/applied.md`).

9.3 Literature search

Agents L (effort xhigh) and L2 (effort medium) searched on 27 October 2026: the primary sources on disk (full texts where stated in `references.md`); OpenAlex and Semantic Scholar citing works of the source, of Blanchini–Giordano, of Craciun's preprint, of Wiuf's preprint and of Conradi–Flockerzi; the arXiv API on "reverse-dual", "equal conflict", "dual reaction network", "transposed stoichiometric", siphon and duality, "free-choice" and reaction, drain and persistence, "global attractor conjecture", "preclusters"; web searches on the same phrases; and the AI campaigns of the project protocol (the Palomar and Evidence Press repositories, the OpenAI mathematics repository, Zenodo, GitHub search). Google Scholar was not reachable. "Not found" in Table 1 refers to this search. No work cites or answers Conjecture 22 or Question 24 apart from this project.

9.4 Credits

Research agents: P (Petri dictionary), E and E2 (exhaustive computations), Q1 (persistence in rank one, drain sets), Q2 (corrected preclusters), Q3 (equal-conflict theory), D2 (rank two): Claude Opus 5.5, effort xhigh; G (non-self-duality, codimension, boundary dictionary) and A2 (Theorems A, AF, O; Theorem 47): GPT-6-Astra, effort xhigh. Literature: L, Claude Opus 5.5, effort xhigh; L2, Claude Opus 5.5, effort medium. Verification: V1, V2, V3, Claude Opus 5.5, effort xhigh; archived report: GPT-6-

Astra, effort xhigh. Writing of this manuscript: Claude Opus 5.5, effort xhigh. The changes made in writing (merged statements, new elementary proofs) are listed in `gac-duality/writer/CHANGES.md`.

Table 76 / Concordance with the working notes in `gac-duality/` (P = `petri/`, E = `computation/`, G = `gac/`, Q1 = `round2/persistence/`, Q2 = `round2/preclusters/`, Q3 = `round2/ec/`, E2 = `round2/computation/`, D2 = `round3/dichotomy/`, A2 = `round3/dichotomy-astra/`).

Here	Notes	Verifier report
2, 3, 6	archive Lemma 2; P 1.1, 1.2	V1 §1.1–1.2
7	P Lemma SC	V1 §1.9, V2 §1
8, 9, 11	archive Prop. 3, 6, Thm 7, Cor. 8–10, Thm 1	archived verification
12	P Rem. 2.5; Q2 Prop. 3.2	V1 §1.4, §2.4
14, 15	Q2 Lemma 1.2, Props 2.1, 2.2	V1 §2.2–2.3
16	Q2 Thm 3.1, Props 3.2, 3.3	V1 §2.4
17	Q2 Thms 4.1, 4.2; E2 Prop. E2.6	V1 §2.5, §3
18	Q2 Prop. 5.1, Cors 5.2, 5.3	V1 §2.6
19	Q2 Thm 6.1, Cors 6.2, 6.3, Prop. 6.4	V1 §2.7
20, 21	P Thms A, B	V1 §1.6, §1.7
22, 23	P Props H, G	V1 §1.14
24	P Thms D, E	V1 §1.10, §1.12
25	P K8; Q3 K8s; E2	V1 §1.11, V2 §2.11
27, 28, 29	P Lemma CF; Q3 Lemmas CF′, M, R	V2 §1
30, 31, 32	P Thm K5, C, C′; Q3 K5gen, C, C^{gen}, CP	V2 §2.1–2.3, §3
33, 34	P Thm T, Cor. T*	V2 §3
35, 37	Q3 Thms U, GS	V2 §2.4, §2.5
38, 39, 40, 41	Q3 Thms T*′, U*, Props GS*≤2, tor*, T*-cex, T*-dim	V2 §2.6–2.9
42, 43	Q3 Thm PC, Cor. BH	V2 §2.10
44, 45	Q3 §7.2, Conj. CR	V2 §2.12
47	E2 Prop. E2.5, Cor. E2.5′; A2 Thms E, E+	V2 §4, V3 §6
48, 49	G Props 2.1, 2.2, §5; E §5	V3 §1–2
50, 51	G §3; E Prop. E.4	V3 §1–2
52, 53	Q1 Thm D, Example C; E; E2	V3 §2–4
54	A2 Thm O	V3 §6
57, 58, 59, 60, 61, 62	Q1 Lemmas 1.1–1.2, Thm S1, S1-N, S1-WR, C-pure, C-2, examples; E2 Prop. E2.1	V3 §3–4
63	D2 Thm W	V3 §5
64, 65, 66, 67	D2 Lemma A, Lemma BM, Thm P2, Props E, H, CM, L, S, B, Example U; E2 Prop. E2.7	V3 §4–5
68, 70, 71	A2 Thms A, AF, Prop. L	V3 §6
72, 73, 74, 75	G Props 2.1, 4.1, 6.1–6.3, Example 8.1; L §12	V3 §1

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